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Energy Consumption and Driving-Action Modeling

From Context-Aware Truck Energy Analysis to Driving-Action Prediction

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Abstract

Energy-aware truck operation depends on separating driver-control behavior from the operating context in which it occurs. Road slope, vehicle mass, ambient temperature, speed, and route conditions can dominate measured energy consumption, so raw energy-use comparisons risk treating difficult operating conditions as inefficient driving. A staged telemetry framework is developed to estimate driver-related energy influence under comparable operating conditions and to explore whether the same structure can support future energy-aware driving actions.

The framework has three model stages. A context-only energy model ($M1$) predicts energy consumption from operating-context signals. A control-aware energy model ($M2$) predicts the same target after adding driver-control signals, including accelerator and brake pedal information. The improvement from $M1$ to $M2$ is used to test whether driver-control signals add predictive value beyond operating context. A driving-action model ($M3$) is then trained using residual-filtered driving windows selected with the energy models and predicts future signed longitudinal control actions. It is evaluated both offline and in a software-in-the-loop vehicle simulator, including replay of recorded driving windows and new simulator-native routes.

Adding driver-control signals improved prediction in all primary comparable-condition buckets. In one representative bucket, the 10 s standard deviation of normalized residuals decreased from 2.60 for $M1$ to 1.49 for $M2$. Sensitivity tests show that slope–temperature–mass buckets, 1–5 s signal resolution, sufficient comparable data per bucket, and the availability of speed, slope, and accelerator pedal signals are important for reliable energy modeling. For driving-action prediction, in simulator tests, $M3$ showed plausible energy-saving behavior by reducing accelerator demand earlier and avoiding unnecessary braking before downhill sections, and it completed a new simulator-native route independently.

Context-aware telemetry modeling provides a more defensible basis for driver-related energy analysis than raw energy-consumption comparisons. The framework can identify useful driver-control information and generate plausible driving-action candidates for energy-aware driver support.

Keywords

Battery electric vehicles, truck energy consumption, driver support, context-aware modeling, telematics, machine learning, energy efficiency, Software-in-the-Loop simulation.

Sammanfattning

Energimedveten lastbilsdrift förutsätter att förarens styrbeteende kan särskiljas från den driftkontext där det sker. Vägens lutning, fordonsmassa, omgivningstemperatur, hastighet och ruttförhållanden kan dominera den uppmätta energiförbrukningen, vilket innebär att direkta jämförelser av energianvändning riskerar att tolka krävande driftförhållanden som ineffektiv körning. Ett stegvis telemetriramverk utvecklas för att uppskatta förarrelaterad energipåverkan under jämförbara driftförhållanden samt för att undersöka om samma struktur kan stödja framtida energimedvetna köråtgärder.

Ramverket består av tre modellsteg. En renodlad kontextmodell för energi ($M1$) predikterar energiförbrukningen utifrån signaler som beskriver driftkontexten. En styrmedveten energimodell ($M2$) predikterar samma målvariabel efter att förarens styrsignaler, inklusive information från gas- och bromspedal, har lagts till. Förbättringen från $M1$ till $M2$ används för att undersöka om förarens styrsignaler tillför prediktionsvärde utöver driftkontexten. En köråtgärdsmodell ($M3$) tränas därefter på residualfiltrerade körfönster som valts ut med hjälp av energimodellerna och predikterar framtida signerade longitudinella styråtgärder. Den utvärderas både offline och i en Software-in-the-Loop-fordonssimulator, vilket innefattar både uppspelning av inspelade körfönster och nya simulatorspecifika rutter.

Tillägget av förarens styrsignaler förbättrade prediktionen i samtliga huvudsakliga bucketar med jämförbara förhållanden. I en representativ bucket minskade standardavvikelsen över 10 s för de normaliserade residualerna från 2,60 för $M1$ till 1,49 för $M2$. Känslighetsanalyser visar att bucketindelning baserad på lutning–temperatur–massa, signalupplösning på 1–5 s, tillräckligt med jämförbar data per bucket samt tillgång till signaler för hastighet, lutning och gaspedal är viktiga för tillförlitlig energimodellering. För prediktion av köråtgärder uppvisade $M3$ i simulatorförsök ett rimligt energibesparande beteende genom att minska gaspådraget tidigare och undvika onödig inbromsning före nedförsbackar, samtidigt som modellen självständigt genomförde en ny simulatorspecifik rutt.

Kontextmedveten telemetrimodellering ger en mer välgrundad bas för analys av förarrelaterad energipåverkan än direkta jämförelser av rå energiförbrukning. Ramverket kan identifiera användbar information i förarens styrsignaler och generera rimliga kandidater till köråtgärder för energimedvetet förarstöd.

Nyckelord

Batterielektriska fordon, lastbilars energiförbrukning, förarstöd, kontextmedveten modellering, telematik, maskininlärning, energieffektivitet, Software-in-the-Loop (SIL)-simulering.

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List of Acronyms and Abbreviations

BEV	Battery electric vehicle
CCAP	Cruise Control with Active Prediction
EC	Energy consumption
ECDF	Empirical cumulative distribution function
$M1$	Context-only energy model
$M2$	Control-aware energy model
$M3$	Driving-action model
MAE	Mean absolute error
PID	Proportional–integral–derivative controller
RMSE	Root mean squared error
RQ	Research question
SIL	Software-in-the-loop
VTAB	Virtual truck and bus
XGBoost	Extreme gradient boosting

Chapter 1

Introduction

Energy-efficient vehicle operation depends on how the vehicle is driven under specific operating contexts, including road slope, vehicle mass, ambient temperature, and traffic conditions. This work studies how driver-related influence on truck energy consumption can be estimated from real-world telemetry by decoupling driver behavior from operating context. This chapter first motivates the application setting and formulates the resulting research problem. The remaining sections state the research questions and summarize the methodological orientation and delimitations of the work.

1.1 Motivation

Road freight transport is under pressure to reduce energy use and emissions while maintaining operational reliability. This challenge is also relevant to Scania (TRATON group), where improved use of vehicle data can support more energy-efficient truck operation and future driver-support functions. More efficient driving is also relevant beyond energy consumption, because well-adapted driving that avoids unnecessary harsh acceleration and braking may reduce mechanical stress and is associated with brake wear, tyre wear, gear-shift behavior, and battery operating conditions [1–4]. Modern trucks, including Scania vehicles, generate large amounts of telemetry data, making it possible to study how vehicle operation, energy consumption, operating context, and driver-control actions interact over time.

Battery electric vehicles (BEV) provide the initial application case in this thesis. In BEV trucks, energy consumption directly affects driving range, charging requirements, route planning, and fleet availability. Regenerative braking creates a unique opportunity for efficient driving, because brake pedal usage can influence not only energy consumption but also energy recovery. This BEV setting is therefore a useful starting point for studying driver-related influence, while the broader problem is how such influence can be estimated from telemetry in a defensible way.

One attractive application of such data is driver support. In this thesis, driver support refers mainly to support for longitudinal pedal use: accelerator pedal and brake pedal behavior. These controls are central because they directly shape propulsion, braking, and regenerative energy recovery in BEV operation.

If the data can reveal which pedal-use patterns are associated with lower energy consumption under comparable operating conditions, it may become possible to build support systems that evaluate driving behavior, provide feedback, or recommend more energy-efficient actions. The

aim is to use data-driven modeling to support more energy-efficient decisions, with the driver as the central actor in the system.

1.2 Problem

In observed driving data, measured energy consumption is shaped by several interacting sources of variation as shown in Figure 1.1. Some factors describe the operating conditions under which the vehicle is driven, such as route, traffic, vehicle, and environmental conditions. Others are tied more directly to the driver's control actions, such as accelerator pedal and brake pedal behavior. Additional variables, such as speed, are mixed because they reflect both the driving situation and the driver's response to it. Gear position is selected automatically by the vehicle, although the driver can choose to override it.

In this thesis, operating context refers to the available non-control and mixed operational variables used to describe the conditions under which driver-control actions are observed: road slope, road curvature, ambient temperature, vehicle mass, and speed. Driver control refers to the available direct control signals: accelerator pedal, brake pedal, and auxiliary brake lever positions. Here, auxiliary lever (often called the retarder lever) is a manual driver control for non-friction braking. In BEV trucks, this control is typically used to request regenerative braking and, where available, additional auxiliary braking functions. The lightly colored nodes in Figure 1.1 mark the factors used in this thesis. If these sources of variation in energy consumption are not separated, the resulting evaluation risks confusing difficult operating conditions with inefficient driving.

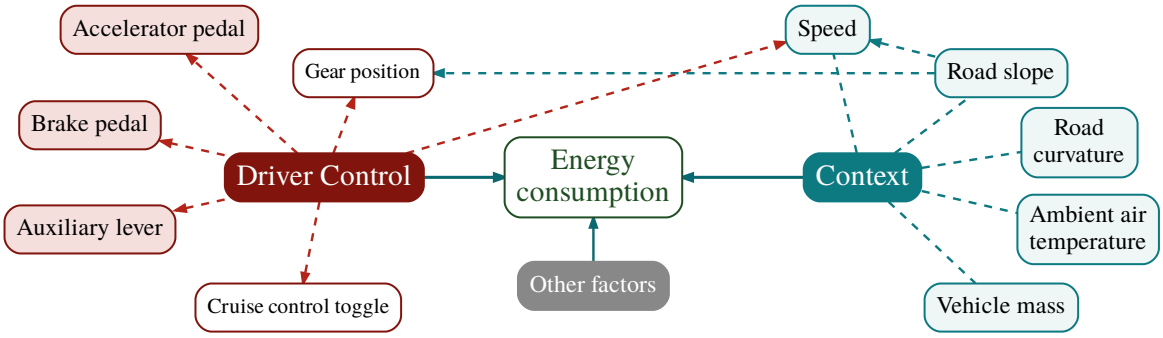


Figure 1.1: Energy consumption (EC) depends on operating-context factors and driver-control signals.

1.2.1 Scientific and Engineering Issues

This problem matters scientifically because it concerns inference under confounding in real-world time series. It also matters in practice because driver-support feedback needs a defensible basis. Telemetry provides measured energy consumption, vehicle state, and driver-control signals, but it does not directly state how much of the observed energy outcome is attributable to context and how much is attributable to the driver. In this thesis, driver-related influence is therefore defined operationally as the additional explanatory value obtained when driver-control signals are added to a model that already accounts for operating context. This is measured by

evaluating a control-aware energy-consumption model against a context-only model under comparable operating conditions. The result is a statistical estimate, not proof that one specific driver action caused one specific energy saving.

The work must handle three related issues. The first is comparability: real-world truck operation spans different routes, vehicle states, vehicle masses, ambient temperatures, and traffic regimes, so evaluation must avoid comparing driver behavior across fundamentally different operating conditions. The second is temporal structure: the meaning of a telemetry row depends on recent signal history, control changes, and segment continuity, not only on the instantaneous values in that row. The third is interpretability: the analysis must make the roles of operating-context variables and driver-control signals explicit enough that any improvement can be interpreted as driver-related information rather than as uncontrolled context.

1.2.2 Research Questions

The original industrial motivation for driver analysis and support is too broad to be treated directly as one scientific problem. The work is therefore structured around three research questions. They separate estimation, robustness, and practical driver-support use.

RQ1. How can driver-related contribution to truck energy consumption be estimated from telemetry data, and how well can the proposed modeling approach distinguish driver-control signals from observable operating context?

RQ2. What data requirements are needed for reliable context-aware energy-consumption modeling, in terms of telemetry resolution, feature availability, and amount of collected driving data?

RQ3. How can the context-aware driver-related influence framework be extended into a driving-support model, and how well can the resulting model predict energy-efficient driver actions and reduce energy consumption in simulator-based evaluation?

1.3 Purpose

The purpose of this degree project is to develop and evaluate a staged data-driven modeling approach for developing energy-aware driving-support in trucks. The work aims to estimate driver-related influence on energy consumption without confusing driver-control behavior with operating context, and to test whether this separation can support a downstream model for energy-efficient driver-control actions.

The expected beneficiaries are vehicle R&D teams, transport operators, and drivers. Vehicle R&D teams can use the framework to evaluate whether available telemetry contains useful driver-control information beyond context. Transport operators may benefit from more defensible tools for energy-efficiency analysis, and drivers may benefit from pedal operation support that is based on situational awareness rather than raw energy consumption.

The work supports sustainability by targeting lower energy use in truck operation, with potential secondary benefits for emissions, component wear, and battery use through well-adapted and energy-efficient driving. The main ethical and social concern is that driver-related feedback can

be misleading or unfair if operating context is ignored. For this reason, the thesis treats driver support as an assistive, driver-centered, and context-aware modeling problem. The aim is to support drivers in making more energy-efficient control decisions, not to rank them universally or replace driver authority with autonomous driving.

1.4 Goals

The goals follow the staged modeling pipeline from data preparation to driving-action evaluation.

1. Build a reproducible workflow for processing truck telemetry and generating features.
2. Define comparable operating conditions for context-aware analysis.
3. Train and evaluate a context-only model for baseline energy consumption.
4. Train and evaluate a control-aware model for the same energy-consumption target.
5. Test whether driver-control signals improve energy-consumption prediction under comparable operating conditions.
6. Assess whether this improvement remains robust across relevant data and modeling choices.
7. Interpret the models to identify influential context and control factors.
8. Explore whether the same framework can support a driving-action model that predicts future driving actions under relevant operating conditions.
9. Evaluate the driving-action model to assess whether its predicted actions are useful.

Figure 1.2 summarizes the model sequence used to support these goals. The first two stages test whether driver-control signals add explanatory value beyond operating context, and the third uses residual-based training examples to predict future driver-control actions.

The main deliverable of the work is a reproducible staged modeling framework that makes the separation between operating context and driver-control signals explicit and testable. The framework connects context-only energy-consumption modeling and control-aware energy-consumption modeling with a downstream driving-action model that uses this separation to predict energy-efficient driver-control actions.

1.5 Research Methodology

This thesis follows an applied empirical and quantitative research approach. It assumes that real-world telemetry can provide useful evidence about relationships between operating context, driver-control signals, and energy consumption, but the evidence must be interpreted cautiously because it is observational, incomplete, and context dependent. The work therefore treats model comparisons as statistical evidence for driver-related information, not as proof of direct causal effects.

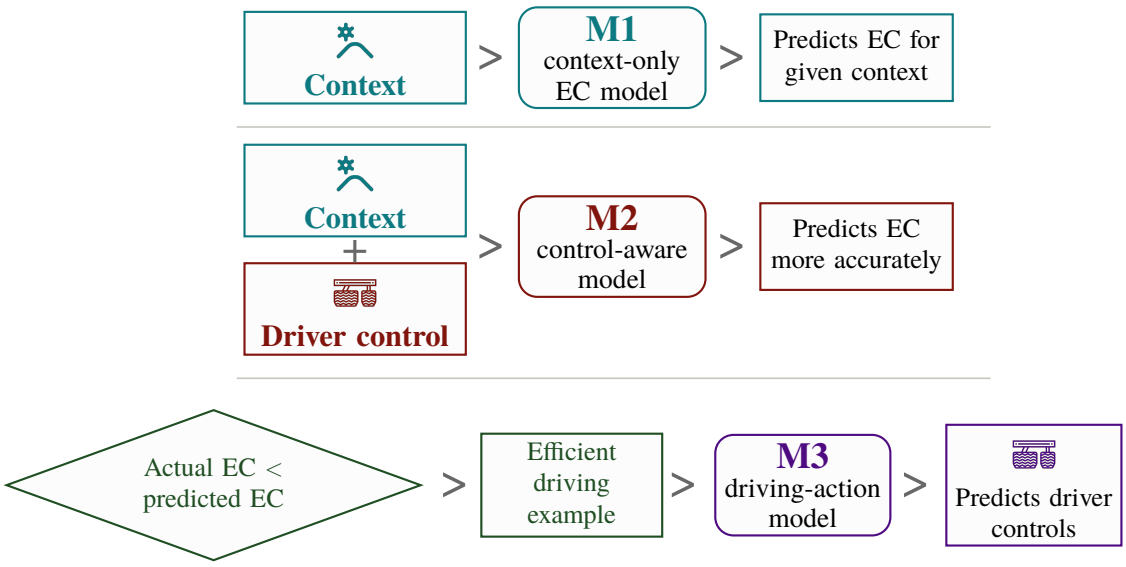


Figure 1.2: Staged model sequence from context-aware energy modeling to driving-action prediction; EC denotes energy consumption.

The chosen method combines telemetry processing, feature generation, supervised learning, grouped model evaluation, and simulator-based driving-action evaluation. This approach was chosen because the research questions require testing whether driver-control signals add predictive information beyond operating context, and whether that information can support a downstream action-prediction model. A purely qualitative method would not answer this predictive modeling question, while a purely physics-based model would not directly test what can be learned from available fleet telemetry. Physics-based modeling could still complement the data-driven approach by adding vehicle-dynamics structure or energy-conservation constraints to improve energy prediction, and fully controlled vehicle experiments could support stronger causal claims; both are treated as outside the scope of this thesis.

The methodology is also engineering-oriented. Data processing, feature design, model selection, experiment tracking, and quality checks are part of the research method because unreliable telemetry handling would directly weaken the scientific conclusion. Chapter 3 gives the detailed methodology for data selection, preprocessing, feature construction, model training, evaluation design, robustness checks, and reproducibility assurance.

1.6 Delimitations

This thesis is limited to offline modeling and evaluation of logged vehicle telemetry. The results apply primarily to the selected BEV truck cohort, with the usable sample further limited by data availability in the chosen comparable-condition buckets. Other truck types, routes, and vehicle configurations are not evaluated as separate study populations.

The work does not produce a universal driver score, and it does not claim to identify the full causal effect of driver behavior on energy consumption. Instead, it tests whether driver-related influence can be estimated more credibly after major operating-context influences have been accounted for. Steering angle was not available in the telemetry data, so the driver-control representation covers longitudinal controls (acceleration and braking), but not lateral control.

Detailed traffic interaction is also outside the reported model scope, although it may affect real driving behavior.

Simulator-based evaluation is used for the driving-action tests. Real-time onboard deployment, production integration, vehicle-level validation, safety-critical control, autonomous driving, replacement of driver authority, fleet-operation optimization, charging strategy, and traffic planning are outside the scope. These areas could be relevant for a deployed driver-support system, but the thesis is limited to testing the proposed staged modeling framework and its offline and simulator evidence.

1.7 Structure of the Thesis

Chapter 2 provides the scientific and technical background for truck energy consumption, driver-support research, driver-behavior analysis, and context-aware machine learning. Chapter 3 integrates the methodology and experimental setup, including comparable-condition bucket design, feature and model contracts, residual-based training-data selection for driving-action model, prediction horizons, training and evaluation workflows, simulator setup, and practical implementation constraints. Chapter 4 presents the results and discusses their practical interpretation, data requirements, and main evidence boundaries. Chapter 5 concludes the thesis and outlines future work. Appendices A–C provide detailed supporting diagnostics for RQ1–RQ3.

Chapter 2

Background

Understanding driver-related influence on truck energy consumption requires background in vehicle energy use, telemetry-driven time-series analysis, driver-support research, driver-behavior modeling, context-aware learning, and structured prediction models. This chapter introduces these areas and explains how prior work informs the modeling choices used in the work.

2.1 Truck Energy Consumption

As introduced in Chapter 1, truck energy consumption is affected by multiple interacting factors. Prior work on energy-efficient driving treats speed, acceleration, deceleration, braking, route properties, weather, and vehicle state as relevant variables [5]. For electrified vehicles, braking behavior is especially relevant because it affects both dissipated kinetic energy and regenerative recovery. Lyu et al. [6] show in an electric-bus setting that separating stopping and non-stopping intersection scenarios improves energy-use prediction, supporting the need to account for operating context when linking behavior to energy consumption.

Walnum and Simonsen [7] analyze truck fuel consumption using behavioral indicators together with factors related to route and mass. Fan et al. [8] show that adding vehicle-mass information improves truck fuel-consumption estimation. These studies support treating vehicle mass and operating context as central explanatory variables rather than secondary adjustments.

2.1.1 Context and Comparability

Energy-consumption studies therefore need to distinguish operating-context variation from behavioral variation before drawing driver-related conclusions. Higher consumption under steeper roads, colder temperatures, or larger vehicle mass should not be interpreted directly as poorer driving.

Comparable-condition analysis responds to this issue by grouping driving data with similar major context variables, such as road slope, ambient temperature, and vehicle mass. It narrows the analysis to observations that share major operating conditions, while recognizing that not all confounding can be removed. This context-aware comparison is aligned with broader context-aware machine-learning literature.

Messmer et al. [9] define context as the information surrounding a situation that is needed for cor-

rect interpretation. They also argue that machine-learning systems can misinterpret data when contextual information is ignored. Contextual information may be spatial, temporal, or spatio-temporal, which is highly relevant to vehicle telematics. Driving data is time-ordered, and the meaning of a driver's action depends on when and where it occurs.

2.1.2 Telemetry-Driven Analysis

Telemetry streams are not analysis-ready by default. Signals may have different time stamps, gaps may occur, driving can alternate between manual pedal-driven periods and cruise-controlled periods. Continuous trips may need to be split into comparable driving segments.

Telemetry quality is not a preprocessing issue alone. It is also a validity issue. Vehicle-data studies show that GPS and onboard-data sources can differ in sampling behavior and dropout. Low-sampling tracking data may also need reconstruction before representative driving-cycle features can be derived [10, 11]. Missingness patterns and mechanisms can affect downstream analysis [12]. Temporal aggregation and feature availability can therefore change which short-duration control events remain visible, influencing both predictive accuracy and the ability to distinguish driver-related influence from noise. This motivates sensitivity analysis for telemetry resolution and signal-group availability.

The models developed in this work are applied to time-series data. Standard tabular machine-learning methods are generally not designed to consume raw sequential streams directly; temporal structure must be encoded through supervised-feature design [13, 14]. In this thesis, that temporal encoding is implemented through lagged values and rolling-window summaries before model training, with grouped evaluation under comparable operating context

2.2 Related Work Areas

The thesis is positioned at the intersection of three main related work areas. The first concerns eco-driving and driver-support research. The second concerns driver-behavior recognition from time-series data. The third concerns context-aware machine learning.

2.2.1 Eco-Driving and Driver Support

Eco-driving is well established as a potentially important source of efficiency improvement, but its exact scope varies across studies and it is often used as a broad term for energy-efficient driving behaviors, interventions, and support strategies [5, 15]. Strömberg and Karlsson [16] report that eco-driving initiatives can improve knowledge and behavior, but that practical implementation remains challenging. Fors et al. [17] show that truck drivers generally favor simple and clear support information over complex interfaces, and that support systems need to be designed with behavioral compliance in mind. Uetake and Nakai [18] provide evidence that personalized feedback and social comparison can help preserve fuel efficiency and offset declines observed in a commercial fleet control group. Mane et al. [19] similarly link behavioral incentives with fuel-efficient truck driving. Jazairy et al. [15] add an important qualitative perspective by showing that motivation to eco-drive does not always translate into behavior, because operational pressures and practical constraints intervene.

This body of work establishes that support systems can affect driver behavior and thereby energy efficiency. It also shows that a driver support system is only useful if it can generate guidance that is credible, actionable, and relevant to the actual situation. If the system evaluates the driver unfairly by ignoring road difficulty, operating context, or vehicle state, its outputs are unlikely to be trusted or operationally useful.

The important distinction is between descriptive feedback and action-oriented driving-support modeling. Feedback systems may tell a driver how efficiently they have driven or which behavior patterns are undesirable after completion of a trip. In contrast, a driving-support model must work while driving to recommend actions that are appropriate for the current and upcoming operating context. This motivates treating action prediction as a separate step after a defensible energy interpretation has been established.

2.2.2 Driver-Behavior Recognition and Energy Linkage

The driver-behavior literature provides techniques for describing or recognizing driving patterns from time-series data. These techniques often use clustering to group similar behavior, segmentation to divide driving into meaningful episodes, or classification to assign observations to predefined behavior categories.

Studies such as Lee and Jang [20] show that recurring behavior patterns can be extracted from naturalistic driving data. This reinforces the relevance of data-driven behavior modeling for real driving traces.

Medeiros et al. [21, 22] study driver-behavior recognition from streaming vehicle data, with a focus on applying models to incoming telemetry and resource-constrained onboard or edge devices. Fang et al. [23] are especially relevant because their work is centered on trucks and combines driving-behavior recognition with fuel-economy evaluation. Their offline-plus-online framework uses offline analysis to learn behavior categories from historical data and online analysis to recognize behavior in incoming driving data. Ashqar et al. [24] further connect driving behavior with fuel-consumption prediction. These studies are close in spirit to the broader Scania motivation: behavior analysis should eventually support feedback or assistance. However, they do not primarily separate the explanatory value of driver-control signals from the influence of observable operating context.

2.2.3 Context-Aware Learning

Context-aware learning refers to modeling approaches that use information about the situation around an observation when interpreting or predicting that observation. Fazla et al. [25] suggest that context-aware ensembles can model time-series data more robustly across changing conditions. Context-aware modeling has also been used with naturalistic truck-driving data. Tang et al. [26] combine driving, vehicle, weather, and environment data with temporal attention for aggressive-driving identification. Their target is different from driver-related influence estimation, but the underlying idea is relevant: a driving action cannot be interpreted well without the situation around it.

This point is central in truck-energy analysis. The same driver action can have different meaning depending on road slope, vehicle mass, ambient temperature, speed regime, and traffic conditions. If a model ignores this structure, it may learn correlations that are specific to one operating

regime rather than a driver-related effect that remains meaningful across comparable conditions.

2.2.4 Longitudinal Driver-Action Prediction

Recent longitudinal-action studies illustrate the diversity of this modeling space. Cavanini et al. [27] compare regression, shallow and deep neural networks, and least-squares support-vector machines for future driver torque demand, while Zhou et al. [28] predict brake intention and intensity using a combination of classification and recurrent modeling. Noda and Yamasaki [29] examine an LSTM-based accelerator-pedal prediction model and how its learned representations relate to driving situations. Tanaka et al. [30] study accelerator-pedal prediction for a truck driver using clustered driving scenarios. Lin et al. [31] use 1 Hz real truck telemetry and upcoming road grade to predict future velocity and acceleration, but not future driver-control signals.

These studies differ from the present work in target, sampling rate, temporal formulation, or vehicle and data regime; their reported numerical results are therefore not directly comparable. They nevertheless motivate evaluating both model-family choice and the use of operating-context information when predicting longitudinal driver actions.

2.2.5 Positioning of This Thesis

Taken together, the related work motivates a staged formulation rather than a direct driver-ranking or driving style recognition approach. Eco-driving and driver-support studies show that driver behavior matters for efficiency, while driver-behavior recognition studies show that recurring behavior patterns can be learned from telemetry. Driver-behavior studies show that recurring patterns can be extracted from telemetry, while longitudinal-action studies demonstrate that future driver actions can be predicted from vehicle data. Context-aware learning adds the central methodological constraint: energy-related behavior should be interpreted relative to the operating situation in which it occurs.

The present thesis combines these perspectives in a staged modeling framework. First, a context-only energy model estimates the energy consumption expected from the operating conditions. Second, a control-aware model adds observed driver-control signals to assess their additional explanatory value under the same comparable conditions. Finally, residuals from these energy models are used to construct training populations for a driving-action model.

The resulting framework is positioned between retrospective energy and behavior analysis and action-oriented driver support. Prior longitudinal-action studies differ in vehicle type, temporal resolution, prediction horizon, and conditioning information. This thesis therefore evaluates whether comparable-condition energy modeling can support driver-related interpretation and guide the selection of demonstrations for different future driving-action profiles.

2.3 Supervised Regression Methods for Structured Telemetry

The supervised regression methods considered in this thesis are applied to telemetry that has first been transformed from raw sensor streams into structured telemetry features. Raw telemetry ar-

rives as time-stamped sensor and control streams, such as vehicle speed, road slope, road curvature, ambient temperature, accelerator pedal position, and brake pedal position. Standard tabular machine-learning models do not use these streams directly. The signals are instead aligned to a common time basis and converted into row-wise numerical feature vectors.

Structured telemetry features therefore combine the current vehicle state with recent signal history. A feature row can contain current values, lagged values, and time-window summaries, as illustrated in Table 2.1. The exact signal-alignment and filling procedure is defined in Section 3.3.1, and the exact lag/window feature definitions are specified in Section 3.3.3.

Table 2.1: Examples of structured telemetry features.

Feature type	**Example**	**Meaning**
Current value	Road slope at time t	Operating context at the current position
Lagged value	Vehicle speed at time $t - 1$ s	Recent past vehicle state
Time-window summary	Mean accelerator pedal position over previous 5 s	Recent driver-control behavior

The same row-wise representation is used with different supervised targets in the staged modeling framework. Table 2.2 shows the distinction between the energy-consumption target and the later driving-action target.

Table 2.2: Examples of constructed model targets.

Model stage	**Target example**	**Purpose**
$M1$, $M2$	Instantaneous energy consumption at time t	Energy-consumption prediction
$M3$	Accelerator/brake action shortly after time t	Near-future driver-control prediction

With this representation, the energy-consumption stages are supervised regression problems: each input is a structured feature row and the output is a numerical energy-consumption target. The driving-action stage uses the same type of structured input, but predicts future driver-control behavior instead.

Linear regression models provide simple structured-regression baselines, and ridge regression extends this method class by adding regularization to reduce sensitivity to correlated inputs [32]. This makes ridge useful as a stable reference model for the telemetry features. At the same time, comparing it with nonlinear supervised regression methods is important because energy consumption can depend on interactions between context, vehicle state, and driver-control signals.

Tree-based ensemble methods are more flexible for heterogeneous tabular data. A single decision tree can represent nonlinear threshold behavior, but its fitted structure can be sensitive to the training data and may not capture all useful interactions. Ensembles address this by combining many trees, so that the final prediction depends on the collective model rather than on one tree.

Random forests train many randomized decision trees and aggregate their outputs [33]. For regression, this aggregation is usually an average of the individual tree predictions; for classification, it is commonly a vote. Figure 2.1 shows the basic structure. The trees are trained independently, which distinguishes random forests from boosting.

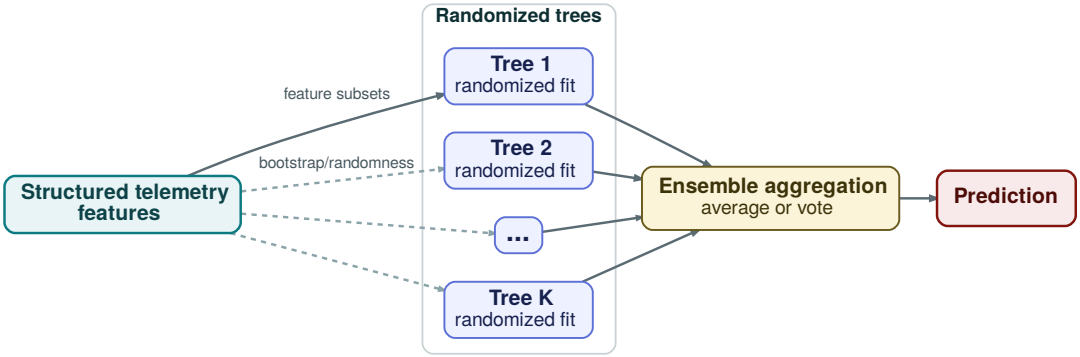


Figure 2.1: Conceptual schematic of a random-forest ensemble, based on Breiman [33].

Boosting combines trees differently. Instead of training independent trees, gradient boosting builds an additive model sequentially: an initial prediction is updated by adding new trees that focus on the remaining prediction error [34]. XGBoost [35] is a widely used implementation of gradient-boosted trees for structured regression and classification tasks. It extends this boosting structure with regularization, scalable tree construction, sparse-feature handling, and efficient approximate split-finding. These properties are relevant for telemetry-derived features because the data can contain nonlinear interactions between operating context, vehicle state, and driver-control behavior, and not all feature groups are equally available or dense. Figure 2.2 illustrates the additive boosted-tree structure. The trees in this figure are sequential correction trees in one boosted ensemble, not independently aggregated random-forest trees.

Key XGBoost hyperparameters have clear modeling roles. The number of estimators controls how many boosted trees are added to the ensemble, while maximum depth controls how complex each tree can become. The learning rate scales the contribution of each new tree, so smaller values typically require more trees but can improve stability. Row and column subsampling control how much of the training data and feature set are used per tree, which helps reduce overfitting and increase model diversity. The regularization coefficients α (L1) and λ (L2) penalize overly complex solutions and improve robustness on noisy tabular telemetry data.

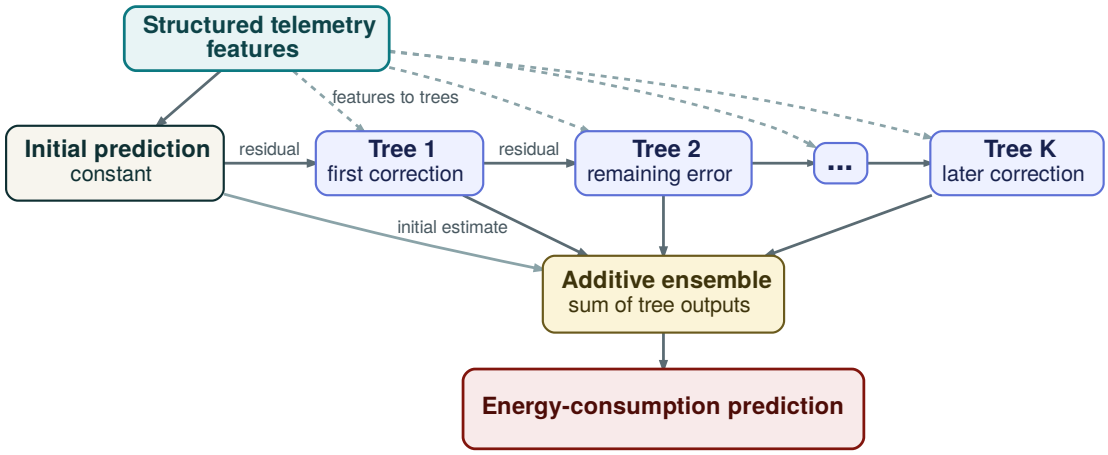


Figure 2.2: Additive boosted-tree prediction structure.

Histogram-based gradient boosting methods, such as LightGBM, use histogram approximations to improve training efficiency on large structured datasets [36]. Feed-forward neural networks are another nonlinear supervised regression method class; with sufficiently many hidden units, multilayer feed-forward networks can approximate broad classes of functions [37]. In this thesis, this class is represented by both a single-hidden-layer wide neural-network baseline and deeper multilayer perceptrons (MLPs). The term "wide" describes the baseline architecture size rather than a separate neural-network type. Neural-network baselines provide a useful predictive benchmark, but their performance can be more sensitive to architectural and optimization choices than that of the tree-based methods considered here. These differences motivate comparing several supervised regression methods before selecting the model type used for the main analysis. Section 3.6.4 describes how these methods are used and selected in the work, and Section 4.2.5 reports the empirical comparison.

2.4 Summary

This chapter has reviewed the background needed for the methodological design. Prior work shows that truck energy use is shaped by vehicle, route, environment, and driver behavior; that telemetry analysis must handle time-series structure and data quality; and that driver behavior can be recognized and supported, but should be interpreted in context when energy consumption is the target. The supervised-regression literature further motivates comparing transparent baselines with nonlinear tabular methods for the structured telemetry features used here.

The next chapter describes how these background ideas are operationalized in the data selection, feature construction, model training, and evaluation design.

Chapter 3

Methodology and Experimental Setup

This chapter integrates the study methodology with its experimental setup. It presents how the staged, context-aware framework is translated into a reproducible telemetry-processing and modeling workflow for estimating driver-related influence under real-world constraints and is consolidated in Table 3.6 placed at the end of the chapter. It is aligned with the research questions and prepares the results in Chapter 4: the $M1$–$M2$ comparison targets driver-related influence, robustness studies address sensitivity to practical data constraints, and the $M3$ analyses cover driving-action prediction and simulator-based evaluation.

The design choices are also shaped by implementation constraints that affect scientific validity, including sparse usable data in comparable-condition buckets, bucket-definition sensitivity, the need for directly measurable targets, and the limits of validating driving-action models on logged data alone.

3.1 Methodological Approach

This work uses an empirical and quantitative method based on logged vehicle telemetry. The method follows the staged problem formulation introduced in Section 1.4. First, a context-only model estimates energy consumption from operating conditions. Second, a control-aware model estimates the same after adding driver-control signals. Third, the validated structure is extended toward short-horizon driving-action prediction.

The data comes from real-world operation rather than from controlled experiments. The method therefore does not claim strict causal effects. Instead, it aims to make the comparison defensible by controlling major operating-context factors, keeping related rows together in grouped splits, repeating experiments across seeds, and comparing models that differ only in the information made available to them.

3.2 Data Source, Study Scope, and Analytical Sample

The data used in the thesis consists of truck telemetry from Scania data sources. The available signals describe vehicle motion, operating conditions, and driver controls over time. The study is carried out offline on logged data; it does not evaluate a deployed real-time system.

The practical study scope is deliberately narrower than the full telemetry population. The thesis focuses on a selected comparable cohort of BEV trucks and on comparable operating conditions in which energy-consumption comparison is meaningful enough to support context-aware interpretation. This means that the data is not treated as one global homogeneous pool. Instead, the analytical scope is defined by the selected vehicle cohort, available telemetry coverage, and comparable-condition buckets.

The driver-control representation is also limited by signal availability. Longitudinal control signals are available and form the basis for the control-aware and driving-action models. Steering angle is not available in the thesis data and is therefore not modeled directly. This limitation is important for interpreting the scope of the driver-related conclusions: the thesis primarily studies longitudinal control (acceleration and braking) behavior rather than complete driver behavior.

3.2.1 Vehicle Cohort Selection and Operational Data Scope

Vehicle selection is purposive rather than purely random. The cohort is defined to balance three requirements: comparable truck configuration, sufficient telemetry coverage, and relevance to the operational focus of the study. The grouping logic was also informed by informal engineering input from Scania test drivers, who emphasized that motor power and overall vehicle layout (rigid or articulated body and trailer attachment) are more influential for perceived driving adaptation than internal configuration details such as axle ratio or gearbox variant. The selection therefore prioritizes driver-facing similarity: BEV propulsion, motor power class, wheel configuration, and chassis adaptation.

A very narrow fixed-truck scope was considered first to maximize similarity, but it became too small after road-slope, ambient-temperature, and vehicle-mass filtering, leaving several buckets with too few usable pedal-driving windows for stable training and evaluation. The reported setup therefore uses a broader but still comparable cohort: 32 BEV trucks with a 6x2*4 layout, non-articulated vehicle type, and a 400 kW electric motor. Across the processed pedal-driving files, this cohort provides approximately 9 169 hours of pedal-driving data, which is sufficient for the $M1$–$M2$ comparison under comparable-condition bucketing.

At the same time, trucks within the cohort are not physically identical. Differences such as roof height, cab variant, axle ratio, gearbox type, and body/application can still influence energy consumption through aerodynamic drag, driveline efficiency, and duty-cycle variation. These factors are treated as residual within-cohort variation rather than as primary grouping dimensions. This first-stage cohort definition reduces avoidable structural heterogeneity, while operating-context variation is handled later through comparable-condition buckets based on road slope, ambient temperature, and vehicle mass.

The operational data is further restricted to pedal-driving periods, where longitudinal motion is controlled by accelerator and brake pedal actions rather than cruise control. This restriction aligns the dataset with the thesis objective of analyzing driver-control signals and driver-related influence on energy consumption. The resulting analytical sample is therefore smaller than the raw telemetry population but methodologically better matched to the research questions.

3.2.2 Analytical Sample

The effective analytical sample consists of filtered rows, continuous driving chunks, and fixed-duration windows derived from those chunks. The number of usable rows and windows depends on the selected vehicle cohort, bucket definition, feature set, and minimum-coverage rules. There is therefore no single sample-size number that describes the whole thesis. The concrete row counts, bucket hours, and model-specific evaluation sizes are reported with the corresponding experimental setup and result tables in later chapters.

Window-level evaluation is used when energy-consumption behavior over a short driving stretch is more meaningful than a single 1 s row. For a window length w, the window-averaged observed and predicted energy-consumption values are defined as

$$\bar{y}_{i,w} = \frac{1}{w} \sum_{j=i}^{i+w-1} y_j, \qquad \bar{\hat{y}}_{i,w} = \frac{1}{w} \sum_{j=i}^{i+w-1} \hat{y}_j,$$

where i is the first row in the window. This notation is used later for 10 s, 20 s, and 60 s window-level comparisons.

The target population of interest is battery electric truck driving under real-world conditions where driver actions may influence energy consumption. The thesis does not claim to represent all truck types, all control modes, or all routes. It evaluates whether the proposed methodology is useful within the defined study scope.

3.3 Data Preparation Pipeline

The telemetry used in the thesis is not analysis-ready in raw form. Signals may arrive at different rates, timestamps may not align, missing values may occur for different physical reasons, and long operational sequences need to be converted into comparable analytical units. A substantial part of the methodology is therefore devoted to transforming raw telemetry into modeling-ready data while preserving temporal meaning.

3.3.1 Signal Alignment, Cleaning, and Filling

The first preparation stage aligns the selected signals onto a common time basis and pivots the data into a row-wise form suitable for later processing. Missing values are not handled with a single generic rule. Instead, the filling logic depends on the physical meaning and expected temporal behavior of each signal. Slowly varying quantities can tolerate different interpolation or carry-forward treatment than quantities whose meaning changes rapidly with motion or route.

This signal-aware treatment is important for methodological validity. A context variable such as ambient temperature can often be handled differently from a variable such as road slope or a direct driver-control signal. The objective is not to create an artificially perfect dataset, but to avoid preprocessing choices that would create clearly implausible behavior in the time series.

The cleaning strategy therefore deliberately favors validity over maximum row count. Long gaps and irregular signal availability are not filled simply to preserve sample size, because doing so would risk creating artificial continuity in variables that are important for energy-consumption

interpretation. This choice reduces the amount of usable data, but makes the remaining chunks more defensible for later model comparison.

3.3.2 Driving-Session and Chunk Construction

After signal alignment and filling, the telemetry is segmented into usable driving periods. The method further refines these periods into continuous pedal-driving chunks. These chunks form an important analytical unit because they preserve local behavioral continuity while making the data manageable for feature generation and grouped evaluation.

The chunk concept is methodologically important for two reasons. First, to not mix clearly unrelated rows into the same analytical unit. Second, closely related rows from the same chunk should not be split carelessly between training and evaluation, because that would create leakage and overstate model performance.

3.3.3 Feature Generation

Feature generation transforms the prepared time-series telemetry into engineered feature vectors that can be used by tabular regression based models, $M1$, $M2$, and $M3$ described in Section 3.5. This raw-stream-to-feature-vector pattern is common in applied time-series learning when models do not consume sequences directly [13, 14]. In this thesis, temporal structure is therefore encoded explicitly through lagged and rolling-window features, while the exact lag, lead, and rolling-summary settings are design choices for the present truck-telemetry setting, informed by explicit temporal encoding and context-aware modeling literature [9, 13, 14].

The feature-extraction procedure is deterministic and stage-wise. After common-time alignment and signal-aware filling (Section 3.3.1), telemetry is split into continuous pedal-driving chunks (Section 3.3.2). For each signal $x(t)$, the pipeline then computes current-value features, lag features (for example $x(t-1)$ and $x(t-10)$), and rolling-window summaries over windows such as 5, 10, and 20 s. The rolling summaries include mean, standard deviation, and local temporal trend. Here, trend or slope denotes the temporal slope of a signal over a window, not the road-slope signal itself. The exact lag, lead, and rolling-summary definitions (including selected window lengths and horizons) are design choices of this thesis for the present truck-telemetry setting, while following the general feature-engineering pattern established in prior time-series and map-supported driving literature.

The feature groups follow the staged model design and leakage-safe information boundaries. For the energy consumption models $M1$ and $M2$, predictors at time t use only information available at or before t. $M1$ uses operating-context features, while $M2$ adds driver-control signals under the same timing constraint. For the driving-action model $M3$, the target is future driver action, so route-preview features (upcoming road slope and curvature) and short-horizon lead summaries are included by design. This is consistent with map-supported driving systems that use route-ahead road information (including geometry and slope-related attributes) for planning and assistance [38, 39]. This feature-design structure keeps the pipeline interpretable and reproducible while remaining practically grounded in information channels available in modern map-supported assistance stacks.

This design also reflects a structural limitation of the XGBoost regressor based driving-action model: it maps feature vectors to outputs but does not contain an internal trajectory planner

or control objective that specifies a desired target speed over time. Without such an inherent planning mechanism, the model would otherwise see only current and lagged state. Hence, $M3$ also uses preview-oriented road-slope and curvature features up to 20 s ahead, vehicle-speed lead and delta information at 20 s, where the delta feature is defined as $\Delta v_{20}(t) = v(t+20) - v(t)$, and distance-to-vehicle-ahead summaries over 1, 10, and 20 s windows. The 20 s speed lead and speed-delta features therefore provide an explicit proxy for short-horizon speed intent.

Figure 3.1 summarizes how the usable feature set is narrowed. The full factor space affecting energy consumption is larger than the telemetry available for modeling. The method therefore assumes that the selected context and driver-control signals capture the dominant measurable variation needed for the $M1$–$M2$ comparison. The feature set is intentionally limited to high-impact and interpretable signals. This keeps the model comparison stable, explainable, and feasible across repeated bucketed experiments. Unobserved or unmodeled factors remain as residual variation, while vehicle-configuration effects are partly handled through cohort selection.

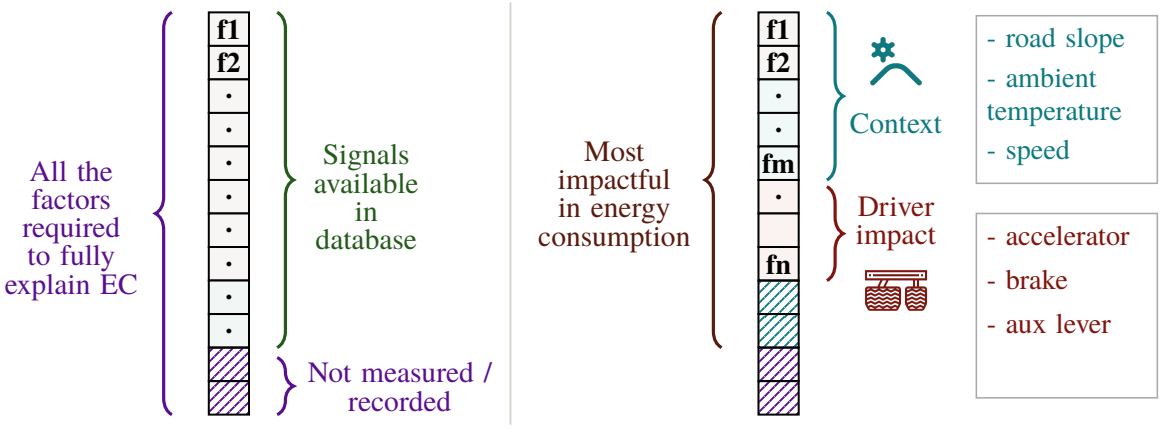


Figure 3.1: Feature-selection logic for context and driver-control modeling.

The overall preparation flow is illustrated in Figure 3.2. The main purpose of this pipeline is to transform raw logged telemetry into comparable pedal-driving chunks and then into reusable feature representations that can support fair model comparison.

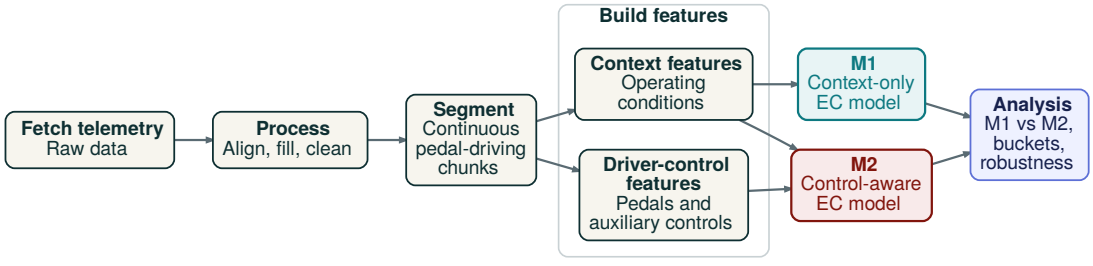


Figure 3.2: Telemetry-to-modeling pipeline.

3.4 Context Definition and Comparable-Condition Bucketing

To compare driving behavior fairly, each telemetry row is interpreted together with its operating context. In this thesis, the context at time t is

$$c_t = [s_t, \tau_t, m_t],$$

where s_t is road slope, τ_t is ambient temperature, and m_t is vehicle mass. These variables are used because they strongly affect energy consumption and can confound driver-related interpretation. The purpose is to make the later $M1$–$M2$ comparison occur within operating regimes that are similar enough to support interpretation.

3.4.1 Why Comparable-Condition Buckets Are Needed

The goal is to compare $M1$ and $M2$ within similar operating context, not across fundamentally different regimes. Bucketing is therefore used as a practical context-control step: it reduces clearly unfair comparisons, even though it cannot remove all confounding. This follows the context-aware principle that observations should be interpreted together with relevant operating context [9].

3.4.2 Bucket Variables and Final Bucket Design

The bucket design uses three contextual dimensions: road slope, ambient temperature, and vehicle mass. These variables were chosen because they strongly influence truck energy consumption and can confound driver-related interpretation.

Bucket assignment is rule-based. Each dimension in $c_t = [s_t, \tau_t, m_t]$ is partitioned into predefined ranges, and each row is assigned to the unique range combination that contains its current c_t. Thus, if the current c_t falls within one slope range, one temperature range, and one mass range, the row belongs to that corresponding bucket. This is interval-membership assignment, not centroid-based clustering (for example, not k-means).

The range limits were not fixed from literature values alone. Initial candidates were set from engineering reasoning, and the final bucket family was selected empirically by sensitivity tests that evaluated the tradeoff between context comparability and usable data volume.

Ambient temperature illustrates this process: battery behavior changes with temperature, especially in colder operation [4, 40], so temperature was included as a context dimension, while its final interval limits were chosen from thesis experiments.

3.4.3 Availability Thresholds and Usable Buckets

Not every theoretically meaningful bucket contains enough data to support grouped model training and evaluation. The methodology therefore treats data availability as a formal part of bucket selection. Buckets need enough usable rows and enough chunk coverage to support train, validation, and held-out evaluation without relying on trivial or heavily imbalanced splits.

This means that the final analytical scope of the thesis is shaped jointly by physics-based context reasoning and by empirical availability constraints. A bucket that is conceptually attractive but too sparse to evaluate honestly is not useful for the thesis.

3.4.4 Enabled Comparable-Condition Buckets

Several candidate bucket combinations were tested before fixing the reported setup. In these trials, bucket definitions differed in how they balanced context control against statistical stability. A bucket definition can be physically meaningful but statistically weak if it leaves too few usable windows, while a broad bucket can hide context variation that should have been controlled.

The final retained setup uses road slope (s_t), ambient temperature (τ_t), and vehicle mass (m_t). The primary result uses a fixed vehicle-mass band of 17–21 tonnes, while an additional higher-mass set is retained as a short generalization check. The final setup represents a practical compromise between controlling road slope, ambient temperature, and vehicle mass and retaining enough observations in each enabled bucket. Table 3.1 summarizes the enabled buckets, their numeric ranges, and their approximate available hours in the filtered study scope. To retain enough data for grouped training and evaluation, the reported enabled buckets were restricted to those containing at least 40 comparable hours. The training-volume sensitivity test in Section 4.2.3 indicates that this threshold is a practical lower guideline for the present setup. The table also illustrates the scale of data fragmentation: although the selected cohort contains more than 9 000 hours of pedal-driving data overall, the enabled comparable-condition buckets contain 1 467 hours in the primary 17–21 t range and 283 hours in the higher 35–39 t range after filtering by road slope, ambient temperature, and vehicle mass.

Table 3.1: Enabled comparable-condition buckets in the reported setup. Road slope is reported in percent, ambient temperature in °C, and vehicle mass in tonnes (t).

Slope s_t range	**Temperature τ_t range**	**Mass m_t range**	**Rows**	**Hours**
Down mild (-5 to -1)	Cold (-10 to 0)	17–21 t	447 687	124
Down mild (-5 to -1)	Mild-cold (0 to 10)	17–21 t	540 305	150
Down mild (-5 to -1)	Near-optimal low (10 to 15)	17–21 t	228 162	63
Flat (-1 to 1)	Cold (-10 to 0)	17–21 t	823 900	228
Flat (-1 to 1)	Mild-cold (0 to 10)	17–21 t	1 161 461	322
Flat (-1 to 1)	Near-optimal low (10 to 15)	17–21 t	561 023	155
Flat (-1 to 1)	Near-optimal high (15 to 20)	17–21 t	272 210	75
Up mild (1 to 5)	Cold (-10 to 0)	17–21 t	468 245	130
Up mild (1 to 5)	Mild-cold (0 to 10)	17–21 t	562 356	156
Up mild (1 to 5)	Near-optimal low (10 to 15)	17–21 t	232 914	64
17–21 t buckets total:			**5 298 263**	**1 467**
Down mild (-5 to -1)	Cold (-10 to 0)	35–39 t	145 998	40
Down mild (-5 to -1)	Mild-cold (0 to 10)	35–39 t	153 122	42
Flat (-1 to 1)	Cold (-10 to 0)	35–39 t	268 464	74
Flat (-1 to 1)	Mild-cold (0 to 10)	35–39 t	308 018	85
Up mild (1 to 5)	Mild-cold (0 to 10)	35–39 t	153 404	42
35–39 t buckets total:			**1 029 006**	**283**

3.5 Modeling Framework

A staged modeling framework is used with three conceptually different models. Each stage has a distinct methodological role, and the stages should not be interpreted as interchangeable.

3.5.1 $M1$: Context-Only Energy Model

$M1$ is the context-only energy-consumption model. It estimates expected energy consumption from operating conditions alone. Its inputs describe the driving situation, such as speed regime, road slope, vehicle mass, and ambient temperature. Direct driver-control signals are excluded.

The role of $M1$ is to provide the contextual baseline. It shows how much of the observed energy-consumption variation can be explained before driver-control behavior is added. This baseline is needed because the later stages depend on separating contextual explanation from potentially driver-related explanation.

3.5.2 $M2$: Control-Aware Energy Model

$M2$ predicts the same energy-consumption target as $M1$, but adds driver-control signals on top of the contextual feature set. The purpose of $M2$ is not simply to obtain a better predictive model. Its methodological role is to test whether driver-control signals contribute explanatory value beyond what the contextual baseline already captures.

If $M2$ improves prediction consistently on the same grouped held-out rows, this supports the claim that the telemetry contains usable driver-control signal. If the improvement is weak, inconsistent, or not robust, then moving toward driving-action modeling would be poorly justified. In this sense, the comparison from $M1$ to $M2$ is the main validation step of the thesis.

3.5.3 $M3$: Driving-Action Model

$M3$ is a candidate driving-action model trained on driving windows selected through residual-based filtering of an energy-consumption model. The reported filtering variants use either the context-only model $M1$ or the control-aware model $M2$ as the selection baseline, together with positive or negative residual-selection ranges. $M3$ predicts short-horizon future control signals that may be useful for driver support. Its inputs include current operating context, preview information about upcoming road slope, and recent control history.

$M3$ is built after the $M1$–$M2$ validation. This ordering is important because the earlier stages establish whether the telemetry contains usable driver-control information. The $M1$- and $M2$-based filtering variants are alternative training-data selection schemes for the same driving-action model stage, not separate model stages. $M1$-based filtering identifies windows that perform favorably or unfavorably relative to operating context alone, whereas $M2$-based filtering evaluates residual behavior after driver-control signals have also been included.

Figure 3.3 provides an overview of the $M3$ pipeline. The following paragraphs describe how the selected energy baseline is used to identify candidate windows, construct future signed-longitudinal targets, and evaluate the resulting model.

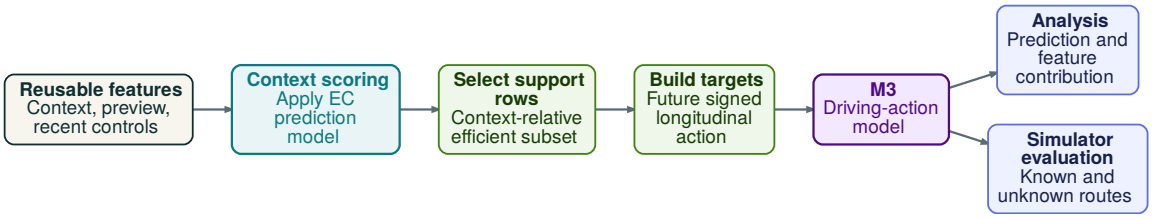


Figure 3.3: Pipeline for constructing, training, and evaluating the driving-action model $M3$.

For a selected energy baseline $B \in \{M1, M2\}$, the baseline model estimates expected energy consumption for the observed context and available input signals. The residuals are aggregated over short selection windows rather than interpreted row by row. Under the telemetry energy-consumption sign convention, a positive residual means that the observed energy consumption is higher than predicted, corresponding to less negative battery discharge or more regeneration. Positive and negative residual-selection ranges are therefore used to define alternative candidate datasets. Rows from the selected windows are then used to construct near-future signed-longitudinal action targets. The signed formulation treats propulsion and braking as opposite directions of one longitudinal action, avoiding separate accelerator and brake recommendations.

To avoid leakage, the residual scores used for $M3$ row selection are generated in a split-consistent way for both $M1$- and $M2$-based variants. For each grouped split, the selected baseline model is trained only on the corresponding training groups, and residual-based window selection is performed within that split protocol. Thus, row selection for $M3$ does not use information from rows held out for that split.

Using the window notation from Section 3.2.2, let

$$\mathcal{W}_{B,\mathcal{R}} = \{k \in \mathcal{W} : \rho_k^B \in \mathcal{R}\},$$

where $B \in \{M1, M2\}$ is the selected energy baseline, ρ_k^B is the signed residual-selection score for window k, and $\mathcal{R}$ is the configured positive or negative selection range. The rows inside the selected windows are then used to construct the signed-longitudinal target. Using t as the time index, the signed action at time t is defined as

$$a_t = \begin{cases} u_{\text{acc},t}, & u_{\text{acc},t} > 0, \\ -u_{\text{brake},t}, & u_{\text{brake},t} > 0, \\ 0, & \text{otherwise.} \end{cases}$$

Here $u_{\text{acc},t}$ and $u_{\text{brake},t}$ denote the accelerator and brake control values at time t. Thus, a_t represents the signed longitudinal driver action, with positive values corresponding to propulsion and negative values to braking.

The horizon definitions used in this thesis are:

a_{t+1} (1 s-ahead signed action, primary controller target), a_{t+10} (10 s-ahead signed action, diagnostic),

and the diagnostic next-window mean target

$$\bar{a}_{t+1:t+10} = \frac{1}{10} \sum_{h=1}^{10} a_{t+h}.$$

Accordingly, the 1 s target is used for the primary closed-loop controller, while longer horizons provide diagnostic evidence of predictive structure.

3.5.4 Final Model and Feature Contracts

A residual-label formulation was considered as a way to model driver-related influence directly. This formulation was not retained because the downstream model was weak on high-magnitude events and compressed predictions toward the center of the distribution. The reported setup therefore keeps the core comparison on a directly measured 1 s energy-consumption target, with driver-control contribution inferred from the matched difference between the context-only and control-aware models.

The final implementation uses fixed model and feature contracts for all stages. $M1$ and $M2$ form the core comparative pair because they predict the same directly measured 1 s energy-consumption target on matched held-out rows, while $M3$ predicts future control actions under the validated context-aware framing. Segment-level behavior is evaluated through short-window aggregation of held-out predictions and observations. This same-target $M1$–$M2$ design keeps the driver-related inference anchored to directly measured energy consumption rather than to an intermediate residual label. Table 3.2 summarizes the unified model and signal contracts used in the reported implementation.

Table 3.2: Unified model and signal contracts used in the reported experiments.

Stage	**Target / required signal basis**	**Input parent-signal groups**	**Methodological role**
Preprocessing	Battery current and voltage are required to construct the 1 s energy-consumption target.	Vehicle speed, accelerator pedal, brake pedal, auxiliary brake lever, and cruise-control status	Constructs aligned and engineering-ready telemetry basis for downstream model stages.
$M1$	1 s instantaneous energy consumption	Vehicle speed, road slope, road curvature, ambient temperature, and vehicle mass	Context-only baseline for expected energy consumption under comparable conditions.
$M2$	Same 1 s instantaneous energy-consumption target as $M1$	All $M1$ inputs plus accelerator pedal, brake pedal, and auxiliary brake lever	Tests whether driver-control inputs add explanatory value beyond context alone.
$M3$	Near-future signed-longitudinal action (reported primary target)	Context, preview road information, recent control history, and distance-to-vehicle-ahead summaries	Driving-action stage for support-oriented future-action prediction and downstream simulator evaluation.

Detailed feature-construction mechanics (current values, lags, rolling summaries, preview features, and target construction) are specified in Section 3.3.3. The temporal settings used in the reported implementation are listed in Tables 3.3 and 3.4.

Beyond the shared $M1$/$M2$ contracts and the primary $M3$ target definition, the reported implementation also evaluates several $M3$ input-design variants. Table 3.5 summarizes these variants and their methodological role. These variants complete the modeling-contract definition and motivate the training and evaluation procedures described next in Section 3.6.

Table 3.3: Exact temporal feature settings of $M1$ and $M2$. Time in seconds.

Stage	**Parent signal group**	**Rolling-summary window lengths**	**Lags**
$M1$, $M2$	Speed, road slope, road curvature	1, 10, 20	1, 3, 5, 10, 20
$M1$, $M2$	Ambient temperature	300	–
$M1$, $M2$	Vehicle mass	1	–
$M2$	Accelerator, brake, auxiliary brake lever	1, 5, 10	1, 3, 5, 10

Table 3.4: Exact temporal feature settings of $M3$. Time in seconds.

Parent signal group	**Rolling-summary window lengths**	**Lags**	**Future rolling-summary window lengths**	**Leads / deltas**
Speed	1, 10	1, 2, 3, 4, 5, 10	–	lead: 20; delta: 20
Road slope, road curvature	1, 10, 20	1, 3, 5, 10, 20	10, 20	leads: 1, 2, 3, 4, 5, 10, 20
Ambient temperature	300	–	–	–
Vehicle mass	1	–	–	–
Distance-to-vehicle-ahead (avg, max, min)	1, 10, 20	10, 20	–	–
Accelerator, brake, auxiliary brake lever	1	–	–	–

Table 3.5: $M3$ variants used in the reported experiments.

Variant	Main difference	Purpose
Full $M3$	Context, preview, and recent-control inputs	Main driving-action model used for direct prediction and simulator-oriented evaluation
No-preview $M3$	Preview-derived inputs removed	Tests whether upcoming road-context information contributes to future-control prediction
Recent-controls-only $M3$	Context and preview inputs removed	Tests whether the model is mostly reproducing recent control behavior
VTAB $M3$ controller	1 s signed-longitudinal target only	Simulator-oriented controller model used when a single immediate action command is required
Random-25% $M3$	Random 25% of eligible training rows	Baseline for testing whether residual-guided selection provides a more useful training population
All-data $M3$	All eligible training rows without residual filtering	Baseline for testing the effect of residual-guided selection against unfiltered training

3.6 Training, Split, and Model-Selection Protocol

The training and evaluation setup is designed to prevent leakage, preserve comparability, and support fair comparison between the model stages.

3.6.1 System Training Execution and Run Traceability

The training implementation has three main workflows. Algorithm 1 shows the shared workflow for the energy-consumption models $M1$ and $M2$. Algorithm 2 shows how the $M3$ training rows are built. Algorithm 3 shows how the driving-action model $M3$ is trained. The algorithms focus on the experimental logic: bucket-wise data selection, grouped splitting, model training, and held-out evaluation.

Each completed run stores a reproducibility record containing vehicle scope, bucket definition, input-feature set, target definition, supervised regression method, split configuration, data-scope summary, and evaluation metrics. These records keep every reported figure and table traceable to the exact configuration that produced it.

A run is treated as a distinct experiment whenever any core ingredient changes (model role, vehicle scope, bucket definition, feature/target contract, model settings, split settings, or random seed). This run identity is required to avoid mixing outputs produced under different assumptions and to preserve valid cross-run comparison.

3.6.2 Grouped Split Strategy

The evaluation uses grouped train, validation, and evaluation splits rather than arbitrary row-wise splits. Rows belonging to the same continuous driving chunk are kept together in the same subset

Algorithm 1 Main training logic for the context-only and control-aware energy models.

Require: Selected model role $r \in \{M_1, M_2\}$, vehicle scope, enabled buckets, input-feature definition, supervised regression method

Ensure: Trained energy model and held-out evaluation metrics

Define the common 1 s instantaneous energy-consumption target.
Select the input-feature set for role r.
For $M1$, use context features only; for $M2$, use context and driver-control features.
for each selected comparable-condition bucket **do**
    Load feature rows for the selected vehicles and select the rows that satisfy the current bucket definition.
    Create grouped train, validation, and evaluation splits, keeping chunk groups intact.
    Train the selected supervised regression method on the training split.
    Evaluate prediction error on the held-out evaluation split.
    Store the trained model and held-out predictions for later comparison.
end for

Algorithm 2 Main logic for constructing the driving-action training rows.

Require: Vehicle scope, enabled bucket definitions, trained $M1$ context models, input-feature definition, residual-selection window settings, residual-percent range

Ensure: Bucket-level $M3$ training rows with context-baseline labels and support-relevance flags

Load the processed feature rows required for context scoring, $M3$ inputs, and future-control targets.
Collapse the enabled bucket definitions into the broader temperature–mass buckets used for $M3$ row selection.
for each selected $M3$ temperature–mass bucket **do**
    Identify the corresponding road-slope–ambient-temperature–vehicle-mass buckets for which $M1$ context models exist.
    for each selected vehicle **do**
        Load the vehicle's processed feature rows.
        for each corresponding $M1$ bucket **do**
            Apply the trained $M1$ model to estimate context-only energy consumption for rows in that bucket.
            Compute the energy residual as observed energy consumption minus the $M1$ prediction.
            Normalize the residual by the observed energy-consumption magnitude.
        end for
        Retain rows belonging to the current $M3$ temperature–mass bucket.
    end for
    Combine retained rows across vehicles.
    Aggregate the $M1$ residuals over short selection windows and compute each window's normalized residual percentile.
    Select windows whose normalized-residual percentile falls inside the configured residual-percent range.
    Mark rows belonging to the selected windows as support-relevant $M3$ training rows.
    Store the selected training rows, context-baseline scores, $M3$ input features, and future-control targets.
end for

Algorithm 3 Main training logic for the driving-action model.

Require: Vehicle scope, enabled buckets, $M3$ input-feature definition, $M3$ target definition, target layout, supervised regression method

Ensure: Trained driving-action model and per-target evaluation metrics

Load context, driver-control, driving-action, and $M3$ target definitions.
for each selected comparable-condition bucket **do**
    Load bucket-level $M3$ training rows derived from the processed telemetry and context-baseline window-selection framing.
    Select support-relevant rows using the stored window-selection criterion.
    Resolve $M3$ input features from current context, preview information, and recent control history.
    Define short-horizon future-control targets.
    if the signed-longitudinal target layout is selected **then**
        Replace separate accelerator and brake targets by one signed longitudinal target and retain auxiliary targets.
    end if
    Create grouped train, validation, and optional test splits according to the selected split configuration.
    Train the selected multi-output driving-action model.
    Evaluate aggregate and per-target prediction metrics on the configured evaluation split.
    Store the trained model, target data, and evaluation predictions for later analysis.
end for

because adjacent rows from the same chunk are strongly related, and splitting them across train and evaluation would create overly optimistic results. In the reported experiments, the grouped split proportions are fixed at 60% train, 20% validation, and 20% held-out evaluation.

At minimum, this requires the training and held-out group sets to be disjoint:

$$G_{\text{train}} \cap G_{\text{test}} = \emptyset.$$

The same disjointness logic is applied to validation groups.

Grouped splitting is essential for the $M1$–$M2$ comparison because both models must be evaluated on matched held-out rows. This ensures that observed differences are attributable to model design rather than to different evaluation samples.

3.6.3 Repeated Runs and Seed Robustness

Single train/evaluation splits can be misleading in a bucketed time-series setting. Repeated runs are therefore treated across different grouped splits as part of the methodology rather than as a minor supplementary check. Variation across seeds is informative because it reveals whether the apparent advantage of the control-aware model is robust or whether it depends too strongly on one specific split.

Because the data is observational and bucketed, residual regime variation can remain within buckets. Robust conclusions therefore rely on average behavior across repeated grouped-split seeds, not on one split. The reported $M1$–$M2$ difference is summarized across comparable-condition buckets and seeds using matched held-out normalized residual spread and RMSE as explained in Section 3.7.1.

3.6.4 Supervised Regression Method Comparison and Final Method Choice

The energy-consumption models are not tied to one supervised regression method by assumption. Before interpreting the final $M1$–$M2$ comparison, the study evaluates seven supervised regression methods on the same task: XGBoost, histogram-based gradient boosting (HistGradientBoosting), bagged trees, random forest, a feed-forward single hidden layer wide neural network baseline, a multilayer perceptron (MLP), and ridge regression. This check is needed because the main result should not depend on an arbitrary default model choice. The MLP comparison uses the smaller of the two deeper feed-forward configurations described below; a larger configuration is evaluated separately as a sensitivity check.

After this comparison, the main supervised regression method chosen for the reported $M1$–$M2$ experiments is XGBoost. Any difference between $M1$ and $M2$ can then be attributed to the added driver-control signals, not to a change in model type.

3.6.4.1 Training Workflow

The $M1$ and $M2$ models are trained under one fixed protocol (same filtered data scope, bucket definitions, grouped splits, and feature/target contracts) so supervised-regression-method differences can be compared fairly. XGBoost is used as the common supervised regression method for the main reported $M1$/$M2$ results. For each run and bucket, trained models and held-out predictions are stored for matched residual-comparison analysis. $M3$ uses a separate workflow because its training rows are selected from the context-baseline framing and its target is future control action.

3.6.5 Baselines for Driving-Action Modeling

The driving-action model $M3$ requires a clear baseline for comparison. In short-horizon future-control prediction, a persistence baseline (also called a naive last-value forecast) provides a standard reference in time-series prediction, because recent control behavior is often informative about the near future [14, 41]. The thesis therefore treats persistence-style prediction as the minimum baseline that $M3$ should improve upon in order to justify its role in driving-action modeling.

3.7 Evaluation and Interpretation

The evaluation framework is aligned with the staged modeling logic. Each model stage is judged according to its own purpose, and no single metric is treated as sufficient on its own.

3.7.1 Evaluation of $M1$ and $M2$

The first research question, RQ1, is addressed through the matched held-out comparison between the context-only model $M1$ and the control-aware model $M2$. The two models are trained to predict the 1 s instantaneous energy-consumption target. Root mean squared error (RMSE), mean absolute error (MAE), R^2, and correlation are retained as standard regression diagnostics,

but the central $M1$–$M2$ interpretation relies on normalized residual behavior. Residual behavior over short windows is especially important, because energy use over a driving segment is more meaningful for later support applications than an isolated instantaneous row.

For observed values y_i and predictions $\hat{y}_i$, RMSE and MAE are defined as

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}, \qquad \text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|.$$

For this reason, normalized residual behavior is an important part of the method. For an observed energy-consumption value y_i and prediction $\hat{y}_i$, the residual is defined as $r_i = y_i - \hat{y}_i$ and the normalized residual as

$$\tilde{r}_i = \frac{r_i}{\max(|y_i|, \epsilon)},$$

where $\epsilon = 1$ avoids division by values close to zero. The same residual logic is applied at row level and after aggregating actual and predicted EC over short windows. The reported normalized residual spread is the standard deviation of $\tilde{r}_i$ on the grouped held-out rows or windows. This makes the comparison between $M1$ and $M2$ less dependent on the absolute energy scale of a bucket and highlights whether unexplained variation narrows consistently across buckets and seeds.

Within one run and one bucket, RMSE and MAE are computed as ordinary (unweighted) means on that run's held-out rows. Weighted aggregation is used only when combining metrics across multiple buckets and/or repeated runs for thesis-level summaries. Row-level summaries use held-out row counts as weights:

$$\bar{m}_{\text{row}} = \frac{\sum_j n_j^{(\text{eval})} m_j}{\sum_j n_j^{(\text{eval})}},$$

where m_j is the metric value for bucket/run j and $n_j^{(\text{eval})}$ is its held-out row count. Window-level summaries use window counts as weights:

$$\bar{m}_{\text{win}} = \frac{\sum_j w_j\, m_j}{\sum_j w_j},$$

where w_j is the number of evaluated windows.

The $M1$–$M2$ evaluation definitions above are operationalized as a residual-comparison workflow which provides the core evidence stream used in the reported RQ1 and RQ2 results (Sections 4.1.1–4.2.5).

3.7.2 Evaluation of $M3$

The third research question, RQ3, is addressed by evaluating whether the driving-action model $M3$ provides useful predictive and simulator-level evidence for support-relevant control behavior. $M3$ is evaluated as a short-horizon future-control predictor. The primary target reported in the thesis is the signed longitudinal action, and mean absolute error is used as the main direct-prediction metric because it measures the average distance between predicted and recorded actions in the same signed-control units. The persistence baseline assumes that the future signed

action equals the current signed action, which is a strong reference for short horizons because driver controls often change smoothly.

For signed-longitudinal action prediction, MAE is computed as

$$\text{MAE}_a = \frac{1}{n}\sum_{i=1}^{n}|a_i - \hat{a}_i|,$$

where a_i is the recorded signed action and $\hat{a}_i$ is the predicted signed action.

The methodological question for $M3$ is not only whether it predicts future controls, but whether context and preview information add value beyond recent control history. This is evaluated through feature-ablation variants that remove preview information or retain only recent controls. Direct prediction quality, persistence comparison, and preview/context contribution provide the main offline evidence before the simulator-based energy evaluation.

The driving-action evaluation uses three layers. First, direct prediction evaluates whether $M3$ predicts short-horizon signed-longitudinal actions better than the persistence baseline. Second, an $M3$-via-$M2$ counterfactual diagnostic tests whether $M3$-implied controls appear beneficial under the trained control-aware energy model while keeping the realized context trajectory fixed. Third, simulator evaluation in VTAB tests closed-loop plausibility when speed, position, and future context evolve from the controller's own previous actions.

3.7.3 Counterfactual Evaluation of $M3$

The counterfactual layer is implemented as an $M3$-via-$M2$ diagnostic. $M3$-implied controls are inserted into the control-aware feature representation, future-control-derived features are rebuilt from those predictions, and the rows are scored with the already trained $M2$ model.

This is a surrogate diagnostic, not a physical closed-loop test: it can indicate whether $M3$ controls appear favorable under the learned energy model, but it keeps the realized context trajectory fixed.

3.7.4 Simulator Setup for $M3$ Evaluation

Virtual Truck and Bus (VTAB) [42] is used here as a software-in-the-loop (SIL) evaluation environment. It is an industrial simulation tool developed at Scania in which mechanical vehicle-component submodels are simulated together with real control units. The simulator contains engine, drivetrain, drive-cycle or OuterWorld, restbus, and driver-model components. In VTAB, OuterWorld is the simulator-native route environment that provides map/grade geometry. The driver-model baseline is VTAB's built-in cascaded PID longitudinal controller, used as an internal reference for speed-tracking.

Scania Cruise Control with Active Prediction (CCAP) is used as an industrial reference for preview-based energy-efficient longitudinal control [43]. CCAP is a predictive cruise-control function that uses topographical map data, GPS or route-preview information, and optimized speed and gear strategies to reduce fuel consumption [44]. CCAP is therefore a relevant benchmark for testing whether the learned $M3$ controller can produce plausible energy-aware behavior in a simulator setting.

In this study, $M3$ is evaluated as a closed-loop controller in VTAB rather than as a precomputed action sequence. The simulator advances stepwise, the current vehicle state and route-preview context are read, the corresponding $M3$ feature row is constructed, and the predicted longitudinal action is applied back to the simulator. As a result, speed, power, and energy trajectories evolve from the controller's own previous actions, making this a stronger evaluation layer than offline prediction on fixed logged rows.

The simulator evaluation uses two VTAB setups, described below: recorded-data replay and simulator-native OuterWorld route control.

3.7.4.1 Recorded-Data Replay Setup

If $M3$-recommended controls are followed, vehicle speed changes, the vehicle reaches different road positions at different times, and the experienced slope may no longer match the logged row sequence. A replay setup therefore has to respect road position, not only timestamp order. This setup replays recorded operating context by overriding a flat-road OuterWorld template. It compares $M3$ against replay of recorded driver controls (accelerator and brake pedal positions) and against the VTAB driver-model baseline.

Pure open-loop distance-based replay is not robust by itself. If the simulated vehicle diverges too far from the recorded speed, it may fail to reach later parts of the recorded control sequence. The recorded replay baseline therefore uses recorded controls where possible, with driver-model support where needed to keep the simulated vehicle close to the logged speed and distance progression. Simple PID and driver-model baselines can also be tested on the same replayed context. To remove startup transients, the first $30\,\mathrm{s}$ of each simulation are treated as a warm-up period and excluded from the energy comparison. The post-warm-up comparison therefore begins at $t_0 = 30\,\mathrm{s}$; cumulative energy is reset to zero at this point, while distance-based plots retain the absolute distance axis.

The post-warm-up cumulative energy used for replay comparison is defined as

$$E_{\text{cumulative}}(t) = \sum_{\tau=t_0}^{t} P(\tau)\Delta t, \qquad t_0 = 30\,\mathrm{s},$$

where $P(\tau)$ is the energy-use rate reported by the simulator summary and $\Delta t = 1\,\mathrm{s}$ is the simulator reporting interval used for the discrete accumulation.

3.7.4.2 OuterWorld Route-Controller Setup

The recorded-data replay setup is not suitable for all controller baselines. In particular, CCAP requires route-horizon information that is not available in the same way when replaying one logged driving window. The implementation therefore separates replay testing from the OuterWorld route-controller test that uses VTAB routes directly. This makes the OuterWorld setup the appropriate place to compare $M3$ with CCAP, a simple PID, and the driver-model baseline on the same simulator-native route.

The corresponding $M3$ results are reported in Section 4.3.

3.8 Robustness and Data-Sensitivity Methodology

As discussed in Section 2.1.2, telemetry quality, temporal resolution, feature availability, and the amount of comparable training data affect both predictive performance and interpretability. The second research question, RQ2, therefore treats robustness to practical data constraints as an explicit part of the method rather than as an optional post-hoc check.

3.8.1 Temporal-Resolution Sensitivity

Temporal-resolution sensitivity is evaluated by rerunning the $M1$–$M2$ comparison after reducing the effective feature update rate. This approximates the practical question of whether similar conclusions would remain possible if telemetry were logged or represented more sparsely. The relevant result is not only the absolute performance at each resolution, but also whether the relative improvement from $M1$ to $M2$ remains stable enough to support the same interpretation. The corresponding results are reported in Section 4.2.2.

3.8.2 Feature-Availability Ablation Workflow

Feature availability is evaluated through one-at-a-time ablation of engineered features under the same grouped train/validation/held-out protocol used in the main $M1$–$M2$ comparison. For the energy models, context-feature ablations test sensitivity to speed, road slope, ambient temperature, vehicle mass, and curvature. For the control-aware model $M2$, driver-control ablations test the contribution of accelerator, brake, auxiliary brake, and combined control groups. Performance change after removing each group is used as a direct diagnostic of whether that group contributes useful predictive information in the reported setup.

This workflow also defines the interpretation boundary of the thesis. Signals that are unavailable, unreliable, or too sparse are not treated as neutral omissions; they constrain what can be claimed. In particular, because steering angle is unavailable, driver-related conclusions are limited to measured longitudinal controls. The corresponding ablation results are reported in Section 4.2.4.

3.8.3 Training-Data Volume Sensitivity

Training-data volume sensitivity is evaluated by subsampling the training portion of each primary comparable-condition bucket while keeping the grouped held-out evaluation rows fixed for each seed. This test asks a practical data-collection question: how much comparable driving data is needed before the context-only and control-aware models behave stably enough to be useful? The result estimates the data-volume behavior for the final bucket set and uses repeated seeds to check whether the pattern is stable under different grouped splits and training samples. The corresponding results are reported in Section 4.2.3.

3.8.4 Supervised Regression Method Comparison

Supervised-regression-method robustness is evaluated by rerunning the same prediction tasks under multiple model families while keeping the data scope, grouped split protocol, and evalu-

ation metrics fixed. This tests whether the main interpretation depends strongly on one model family or remains directionally consistent across alternatives. The corresponding results are reported in Section 4.2.5.

3.8.5 Preview- and Context-Feature Contribution Test for $M3$

To test whether $M3$ uses more than short-horizon control persistence, the methodology compares a small set of controlled input-design variants under the same training and grouped evaluation protocol. The full $M3$ variant uses context features, preview features, and recent-control history. A no-preview variant removes route-preview-derived inputs while keeping the remaining inputs unchanged. A recent-controls-only variant removes context and preview inputs, retaining only recent-control features. This variant set isolates whether predictive gains are attributable to preview/context information or to short-term autocorrelation in control signals.

All variants use the same target definition, train/validation/held-out split logic, and evaluation metrics as the main direct-prediction analysis, so differences can be interpreted as feature-group contribution rather than as protocol artifacts. Variant performance is compared primarily through signed-longitudinal MAE across the same grouped held-out rows and horizons. The corresponding results are reported in Section 4.3.2.

3.9 Reliability, Validity, and Methodological Limitations

Reliability and validity are central concerns because the target quantity is not directly observed as a ground-truth driver-efficiency label. The following subsections evaluate method validity, data validity, method reliability, and limits of generalization within this thesis scope.

3.9.1 Method Validity

The validity of the method depends first on whether the staged decomposition is reasonable. The context-only model is valid only to the extent that the selected context variables explain a substantial share of the important non-driver variation in energy consumption. This is why direct driver-control signals are excluded from $M1$ and introduced only in $M2$.

Another validity question concerns the use of context variables that may themselves partly reflect driver behavior, especially speed. This does not invalidate the method automatically, but it does make leakage analysis and interpretability important. Likewise, the buckets are not assumed to eliminate all hidden confounding; they are a pragmatic context-control mechanism rather than a guarantee of causal isolation.

3.9.2 Data Validity

Data validity is improved by signal-aware preprocessing, long-gap handling, and restriction to continuous pedal-driving chunks. These steps reduce clearly implausible artifacts and align the analytical sample with the research question. At the same time, the thesis remains limited by what is observable in the telemetry. Unmeasured factors such as detailed traffic microstructure, driver state, or latent route conditions may still affect the results.

3.9.3 Method Reliability

Method reliability is supported by a standardized pipeline with reusable feature files, saved models, explicit settings, and repeatable experiment configurations. Grouped splits and saved outputs reduce the risk that conclusions are driven by undocumented interactive or descriptive analysis state or accidental leakage.

Repeated runs across seeds also contribute to method reliability by revealing how sensitive the conclusions are to the precise split of the data. A method that performs well only on one convenient split is not reliable enough for the thesis objective.

3.9.4 Implementation Challenges and Mitigations

Three implementation constraints shaped the final pipeline beyond the validity and reliability points above. The first was telemetry synchronization and missing data. The logged signals did not all arrive with the same temporal behavior, and missing intervals could not be treated as harmless gaps. Signal-aware filling, long-gap rejection, and strict continuous-chunk construction were therefore used to avoid creating artificial time-series continuity. This reduced the amount of usable data, but made the retained driving windows more suitable for energy-consumption modeling.

The second constraint was scale. The useful analytical sample required fetching and processing data across a larger comparable cohort, while feature generation added rolling, lagged, and preview-derived columns. To keep this manageable, the implementation used staged processing and bucket-wise or vehicle-wise loading instead of full-cache processing where unnecessary. This made setup slower but kept the workflow tractable for repeated experiments.

The third constraint was feature availability. Some potentially useful signals were unavailable, too sparse, or not reliable enough to use as ordinary continuous predictors. Such signals were therefore either excluded from the reported model contracts, kept as experimental augmentations, or reframed as conditional features rather than added directly. Accordingly, the conclusions should be interpreted as applying to the available longitudinal, route, and vehicle-context telemetry, not to a complete description of driver behavior and surrounding traffic.

3.9.5 Limitations of Generalization

The results apply most directly to the selected truck cohort, the selected pedal-driving periods, the chosen comparable-condition buckets, and the available telemetry signals. Within this scope, the thesis provides proof-of-concept evidence that the proposed context-aware methodology works for estimating driver-related influence from truck telemetry. The same methodology is expected to be transferable to other truck settings, provided that comparable context variables, driver-control signals, and sufficient operating data are available.

3.10 Experimental Design Summary

To provide a concise reference for the results chapter, Table 3.6 summarizes the fixed experimental settings used throughout the reported analyses. It consolidates the data scope, grouped

evaluation protocol, metric aggregation, prediction horizons, and simulator conventions. Chapter 4 follows this design map when presenting the RQ1–RQ3 results.

Table 3.6: Compact map of fixed design settings used in the reported experiments.

Design item	**Reported setting**
Grouped split policy	Chunk-level grouped split with disjoint train/validation/held-out groups (Section 3.6.2).
Train/validation/held-out proportions	60/20/20.
Number of grouped-split seeds	5.
Primary supervised regression method	XGBoost for the main $M1$/$M2$ and $M3$ experiments; alternative methods are evaluated separately for model-family sensitivity (Section 3.6.4).
Primary comparable-condition scope	Road slope–ambient temperature–vehicle mass buckets; primary mass band 17–21 t, with 35–39 t retained as a short generalization check (Section 3.4.4).
Comparable-bucket availability rule	At least 40 comparable hours for the reported enabled buckets; a practical lower guideline supported by the training-volume sensitivity test (Section 3.8.3).
Normalized residual definition	$\tilde{r} = (y - \hat{y}) / \max(\lvert y \rvert, 1)$; normalized residual spread is the standard deviation of $\tilde{r}$ (Section 3.7.1).
Cross-bucket/run aggregation weights	Row-level summaries weighted by held-out row count; window-level summaries weighted by window count (Section 3.7.1).
Reported $M3$ prediction horizons	1 s primary controller target; 10 s point-ahead and next-10 s-mean diagnostic targets (Section 3.5.3).
Candidate residual-percentile ranges for $M3$	Table 4.5.
Simulator warm-up exclusion	First 30 s excluded before replay/OuterWorld energy comparisons (Section 3.7.4.1).

Chapter 4

Results and Discussion

This chapter evaluates the proposed framework against the research questions defined in Chapter 1. The analysis first examines whether driver-control signals add explanatory value beyond operating context, then evaluates the data and modeling requirements needed for this comparison, and finally tests whether the resulting driving-action model can be used in simulator evaluation. Detailed diagnostic tables and trace plots are reported in Appendices A–C.

Table 3.6 summarizes the fixed design settings used in the reported experiments. For the primary $M1$/$M2$ results in Sections 4.1.1–4.2.4, the shared workflow from Section 3.6.4 uses XGBoost as the common supervised regression method. Table 4.1 lists the fixed hyperparameter values for those runs. Supervised-regression-method sensitivity is evaluated separately in Section 4.2.5.

Table 4.1: XGBoost hyperparameters used for the primary reported $M1$ and $M2$ results.

Hyperparameter	**Value**
Number of estimators	400
Learning rate	0.05
Maximum depth	8
Minimum child weight	1.0
Row subsample ratio	0.8
Column subsample ratio per tree	0.8
L1 regularization coefficient (α)	0.0
L2 regularization coefficient (λ)	1.0

4.1 Driver-Related Energy Influence

RQ1 asks whether driver-related contribution to truck energy consumption can be estimated from telemetry after accounting for observable operating context. In this thesis, that question is tested through the staged energy-model comparison introduced earlier, using the comparable-condition buckets defined in Section 3.4.4. The expected interpretation is that, if $M2$ consistently reduces prediction error relative to $M1$ on the same held-out driving data, the added driver-control signals provide evidence of explanatory value beyond the selected context variables. This section first presents the main held-out result, then checks whether the conclusion remains visible across averaging windows, repeated grouped splits, and different operating scenarios.

4.1.1 $M1$–$M2$ Held-Out Comparison

The held-out comparison uses the same evaluation rows for $M1$ and $M2$ within ten 17–21 t comparable-condition buckets. Each bucket groups rows with similar slope, ambient temperature, and vehicle mass.

Two error views are used. RMSE reports prediction error in the original energy-consumption units. Normalized residual spread compares buckets more fairly by relating the remaining prediction error to the natural energy-consumption variation within each bucket. Negative Δ values denote $M2$ minus $M1$, so lower values indicate improvement from adding driver-control signals.

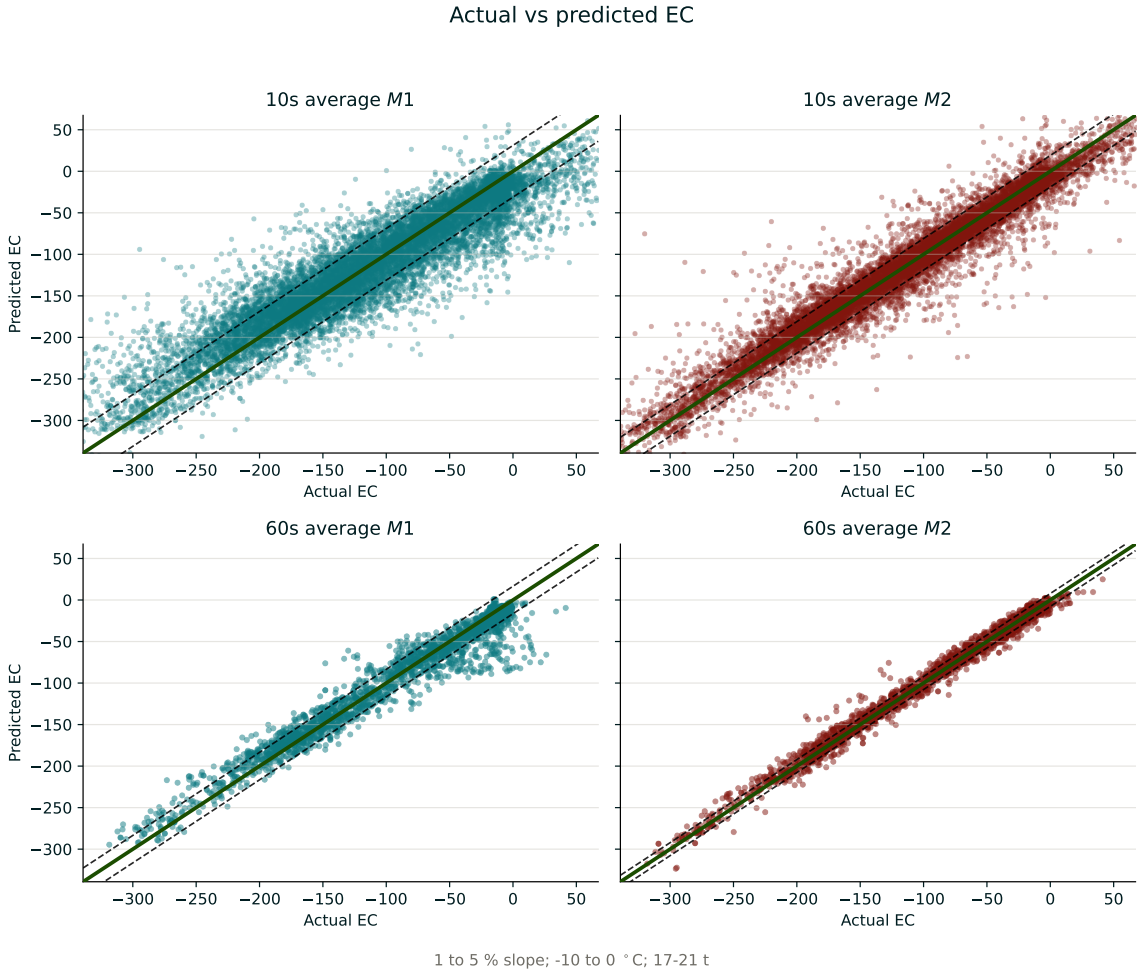


Figure 4.1: Actual and predicted energy consumption at 10 s and 60 s averaging scales.

Figure 4.1 shows the held-out comparison in a selected bucket with slope from 1 to 5 %, ambient temperature from -10 to 0 °C, and vehicle mass from 17 to 21 tonnes. The figure averages actual and predicted energy consumption over 10 s and 60 s windows. These windows are used because segment-level energy consumption is more meaningful for later driver-support use than instantaneous row-level energy consumption, and because short-window averaging reduces high-frequency noise that is less relevant to an energy-efficiency recommendation.

With this window-level view, the $M1$ predictions are more spread around the ideal diagonal, while the $M2$ predictions are pulled closer to the actual energy-consumption values. The raw residual standard deviation decreases from 31.15 to 19.00 at 10 s and from 16.53 to 7.81 at 60 s. In normalized terms, the 10 s residual spread decreases from 2.60 to 1.49, and the 60 s

residual spread decreases from 2.88 to 0.79, showing that the improvement is especially clear when prediction is evaluated over a longer driving segment. The 60 s panels also show that prediction over longer segments can follow the observed energy-consumption pattern, with $M2$ giving the tighter match. A residual-distribution view of the same selected bucket is included in Appendix A.1.

4.1.2 Variation by Operating Scenario

The operating scenario is represented by road-slope–ambient-temperature–vehicle-mass condition buckets. Figure 4.2 separates two effects: the position of the $M1$ point shows the residual spread of the context-only model in each operating scenario, while the horizontal shift from $M1$ to $M2$ shows how much residual spread is reduced when driver-control signals are added. Table 4.2 gives the corresponding RMSE values.

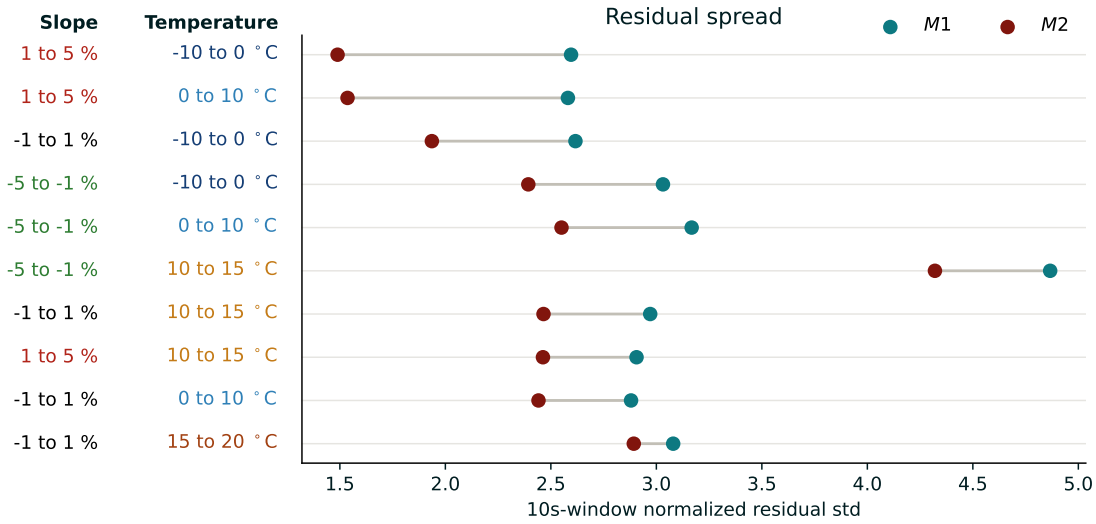


Figure 4.2: Normalized residual standard deviation for $M1$ and $M2$ across the primary 17–21 t comparable-condition buckets.

The residual spread is not uniform across operating scenarios. For the context-only model, the 10 s-window normalized residual spread ranged from 2.58 in the 1 to 5 % slope (mild-uphill), 0 to 10 °C bucket to 4.87 in the −5 to −1 % slope (mild-downhill), 10 to 15 °C bucket. The control-aware model followed the same broad pattern, ranging from 1.49 in the colder mild-uphill, −10 to 0 °C bucket to 4.32 in the same mild-downhill, 10 to 15 °C bucket. Adding driver-control signals therefore improved the model, but it did not remove the underlying scenario dependence of the energy-consumption prediction problem.

The horizontal $M1$–$M2$ shift sizes show where driver-control signals added more or less explanatory value after conditioning on the available context. The largest improvement was in the colder mild-uphill bucket (1 to 5 %, −10 to 0 °C), with a reduction of 1.11. The weakest improvement was in the warm flat bucket (−1 to 1 %, 15 to 20 °C), with a reduction of 0.19.

Table 4.2 reports the same all-bucket comparison in RMSE and 10 s normalized residual spread. The bold row corresponds to the illustrative bucket used in Figure 4.1.

The RMSE values confirm the same directional result: $M2$ reduces error in every primary bucket. The reduction ranged from 8.15 in the colder flat bucket (−1 to 1 %, −10 to 0 °C)

Table 4.2: $M1$–$M2$ comparison on the ten primary 17–21 t comparable-condition buckets.

Slope (%)	Temperature (°C)	Data hours	$M1$ row RMSE	$M2$ row RMSE	Δ row RMSE	$M1$ 10 s norm. res. std	$M2$ 10 s norm. res. std	Δ 10 s norm. res. std
-5 to -1	-10 to 0	124	52.74	39.42	-13.33	3.03	2.39	-0.64
-5 to -1	0 to 10	150	52.74	38.99	-13.75	3.17	2.55	-0.62
-5 to -1	10 to 15	63	51.00	37.26	-13.74	4.87	4.32	-0.55
-1 to 1	-10 to 0	228	49.28	41.13	-8.15	2.62	1.94	-0.68
-1 to 1	0 to 10	322	51.04	38.69	-12.36	2.88	2.44	-0.44
-1 to 1	15 to 20	75	53.60	42.69	-10.91	3.08	2.89	-0.19
-1 to 1	10 to 15	155	50.41	37.90	-12.51	2.97	2.47	-0.51
1 to 5	**-10 to 0**	**130**	**59.05**	**43.54**	**-15.52**	**2.60**	**1.49**	**-1.11**
1 to 5	0 to 10	156	58.66	42.49	-16.17	2.58	1.54	-1.04
1 to 5	10 to 15	64	61.71	47.42	-14.29	2.91	2.46	-0.44

to 16.17 in the mild-uphill (1 to 5 %), 0 to 10 °C bucket. The bucket ordering is not identical to Figure 4.2, because RMSE measures absolute prediction error while normalized residual spread relates the residual variation to the natural energy-consumption variation within each bucket. The common result is that driver-control signals add predictive information beyond the selected context variables in every primary bucket.

In the normalized residual-spread view, the control-aware improvement is therefore most visible in colder uphill operation and weakest in the warmest flat bucket, although the incomplete set of slope–temperature combinations means that this should be read as a descriptive pattern rather than a separate statistical effect. One plausible explanation is that uphill energy use is more directly tied to propulsion demand and accelerator behavior, so driver-control signals can explain a larger share of the remaining variation. In downhill operation, coasting, braking, speed-control behavior, route constraints, and other unobserved states may leave more residual variation even after the available driver-control signals are added. Thus, operating scenario affects both absolute model performance and the amount of additional information gained from driver-control signals.

4.1.3 Robustness and Sanity Checks

Two checks tested whether the main result was sensitive to evaluation scale or grouped train–test split. The comparison was repeated at row, 10 s, 20 s, and 60 s evaluation scales, and across five grouped split seeds. In both cases, $M2$ retained lower residual spread than $M1$, supporting the interpretation that driver-control signals add explanatory value beyond the selected context variables.

A separate sanity check applied the same bucketed comparison to the available 35–39 t buckets. The same direction of improvement was observed, supporting the bucket-based analysis design. The detailed window-level, seed-level, and higher-mass tables are included in Appendix A.1.

4.2 Data Requirements and Robustness

This section identifies the data requirements for the $M1$–$M2$ framework in the tested setup: bucket definition, feature resolution, comparable training-data volume, available signal groups, and supervised regression method choice. The purpose is to show where the framework remains reliable and where performance begins to degrade, so that the results can be translated

into practical guidance for similar telemetry-based studies. The results use the XGBoost configuration listed in Table 4.1. These analyses implement the robustness procedures defined in Sections 3.8.1–3.8.4.

4.2.1 Bucket-Definition Sensitivity and Final Bucket Choice

The final bucket design was selected through sensitivity tests because comparable-condition buckets must control major operating-context variation while retaining enough data for grouped training and evaluation. The study evaluates bucket definitions by absolute $M1$ and $M2$ accuracy rather than by the $M2$–$M1$ improvement alone. This matters because a bucket definition can show a large $M2$–$M1$ difference while still leaving high prediction error if the operating context is poorly controlled.

The bucket-design result has two parts. First, a controlled family-selection test compared buckets based on road slope, ambient temperature, vehicle mass, and their two- and three-way combinations. Second, range-refinement tests checked whether the selected family should use finer slope bins, a split of the 10–20°C temperature range, or a different vehicle-mass width. In all cases, buckets with insufficient data were disabled using an availability threshold, and the plotted values are weighted by the number of available rows in each bucket.

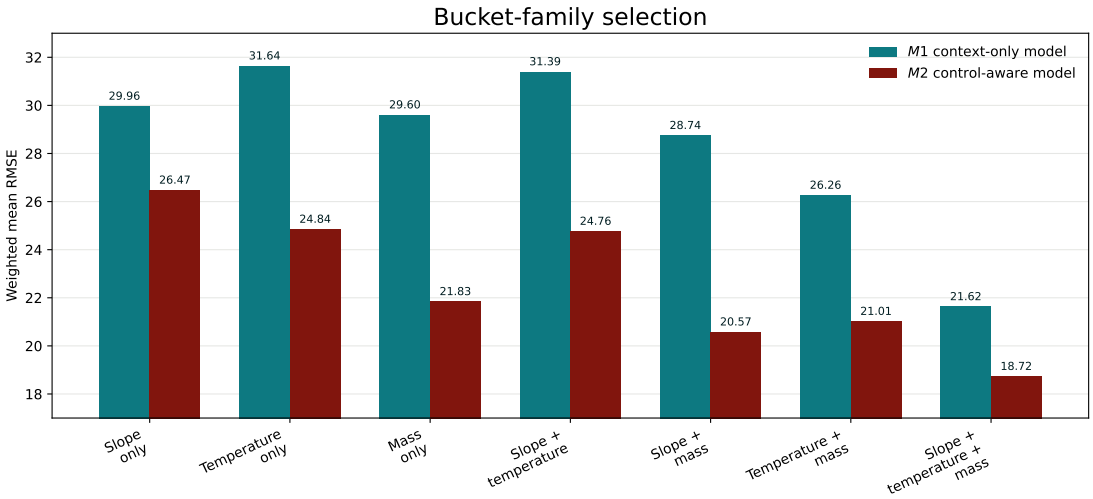


Figure 4.3: Weighted mean RMSE by bucket family.

Figure 4.3 shows that the slope–temperature–mass family had the lowest weighted mean RMSE for both $M1$ and $M2$. The temperature–mass family was the strongest two-dimensional alternative, but adding road slope on top of temperature and mass reduced the remaining error further. The one-dimensional families were clearly weaker, so temperature-only or mass-only buckets were not sufficient for the main comparison. The slope–temperature–mass family was therefore retained for the main analysis.

Figure 4.4 shows the range-refinement decisions after selecting the slope–temperature–mass family. Uniformly finer slope bins slightly increased weighted RMSE, so the coarser slope bins were retained. Splitting the 10–20°C range into 10–15°C and 15–20°C reduced RMSE in the available flat, 17–21 t temperature-refinement test, so this temperature split was kept. For mass, the 17–21 t interval gave the lowest weighted RMSE on common slope-temperature buckets, so it was retained as the primary mass range. The detailed tables are included in Appendix B.1.

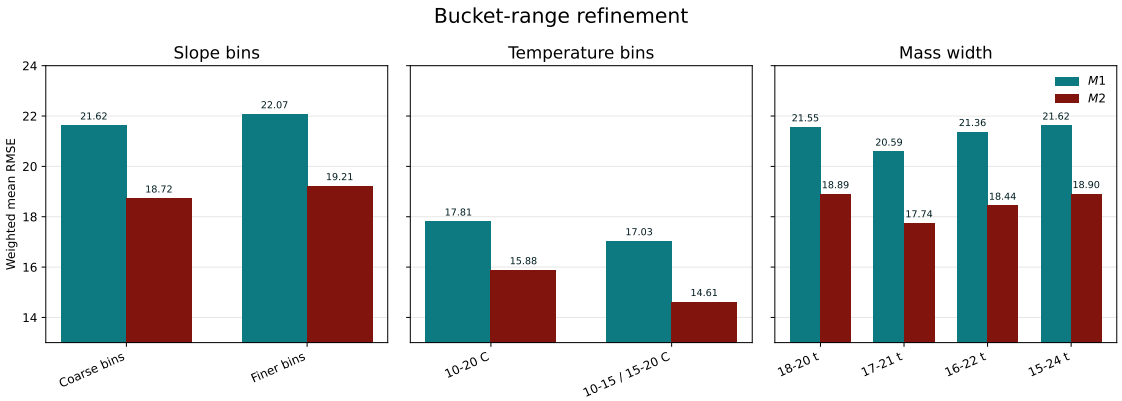


Figure 4.4: Weighted mean RMSE for bucket-range refinement tests.

4.2.2 Temporal-Resolution Sensitivity

The temporal-resolution result tests how much the energy-model comparison changes when the model inputs are made coarser. The same model roles were trained and evaluated after reducing the effective feature resolution from 1 s to 2 s, 5 s, and 10 s for the main comparison. This differs from the window-averaged evaluation in Section 4.1.1: here the input feature table is rebuilt after keeping only rows at coarser effective update intervals, so the result tests data-resolution sensitivity rather than only segment-level reporting.



Figure 4.5: Weighted normalized residual standard deviation for $M1$ and $M2$ under reduced feature resolutions.

Figure 4.5 shows that $M2$ retained lower 10 s-window normalized residual spread than $M1$ at every tested feature resolution. The 1–5 s feature resolutions were similar: $M1$ remained between 2.91 and 3.08, while $M2$ remained between 2.47 and 2.50. The slightly lower $M1$ value at 5 s is interpreted as a small non-monotonic aggregate effect rather than evidence that 5 s resolution is generally better than 1 s resolution. One plausible explanation is that the context-only model is more sensitive to short-timescale variation that it cannot explain directly; the 5 s representation may suppress some of this variation, while $M2$ can already account for part of it through driver-control signals. At 10 s, residual spread increased more clearly, especially for $M1$, which

rose to 3.94 while $M2$ rose to 2.80. The extended 60 s-window check showed the same ordering through 60 s feature resolution, but with clearer degradation at coarser inputs; the detailed values are included in Appendix B.2.

4.2.3 Training-Data Volume Sensitivity

The training-data volume result estimates how much comparable driving data was needed before held-out performance started to stabilize in the selected setup. The comparison was repeated on three 17–21 t buckets with more than 100 comparable hours: mild downhill and cold (-5 to -1 %, -10 to 0 °C), flat and mild-cold (-1 to 1 %, 0 to 10 °C), and mild uphill and cold (1 to 5 %, -10 to 0 °C). For each bucket and each of five grouped split seeds, the held-out evaluation rows were kept fixed while the training rows were subsampled from 10% to 100% of the available training set.

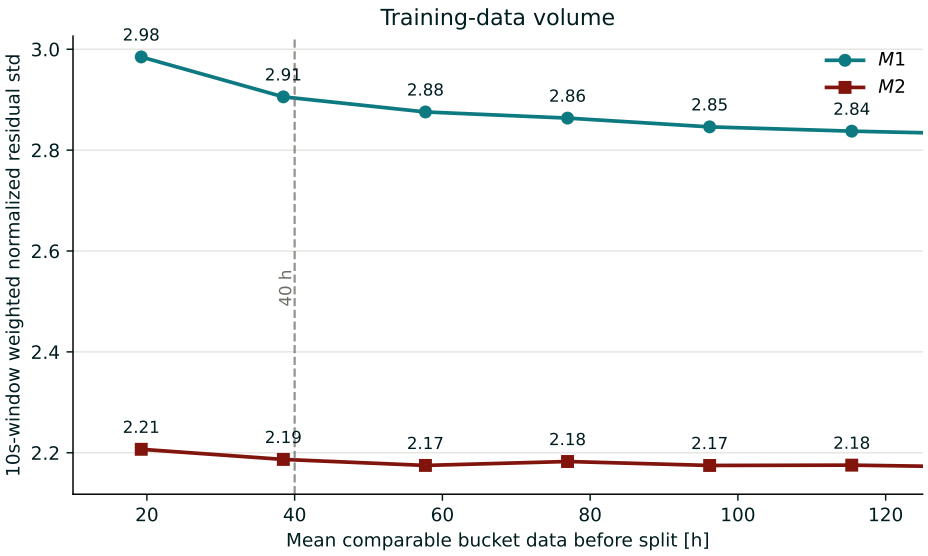


Figure 4.6: Aggregate 10 s-window normalized residual spread across comparable bucket-data volume.

Figure 4.6 shows that both models improved as comparable bucket-data volume increased, using the same 10 s-window normalized residual spread as the primary RQ2 metric. In the aggregate trend over the three tested buckets, a substantial part of the visible improvement was reached by about 40 comparable hours per bucket. With the grouped split used here, this corresponds to roughly 24 hours of sampled training data, because about 60% of the comparable rows are available for training. Additional data still improves some cases, but the aggregate gains become smaller. Table 4.3 gives the corresponding aggregate values for selected training fractions; $M2$ remained better than $M1$ at every tested training size.

4.2.4 Feature-Availability and Signal-Group Ablation

Feature-group ablation identifies which available signals the context-only and control-aware results depend on. This test used all ten primary 17–21 t comparable-condition buckets. One feature group was removed at a time, and the resulting 10 s-window normalized residual spread

Table 4.3: Selected training-data volume sensitivity results.

Training fraction	Mean training hours	$M1$ 10 s weighted norm. res. std	$M2$ 10 s weighted norm. res. std	$M1$ weighted row RMSE	$M2$ weighted row RMSE
0.10	11.47	2.98	2.21	56.56	42.00
0.20	22.93	2.91	2.19	55.22	41.60
0.50	57.33	2.85	2.17	54.08	41.22
0.70	80.26	2.83	2.17	53.79	41.12
1.00	114.65	2.82	2.16	53.55	40.99

was compared with the corresponding full model. Positive changes therefore mean that removing the feature group made the model worse.

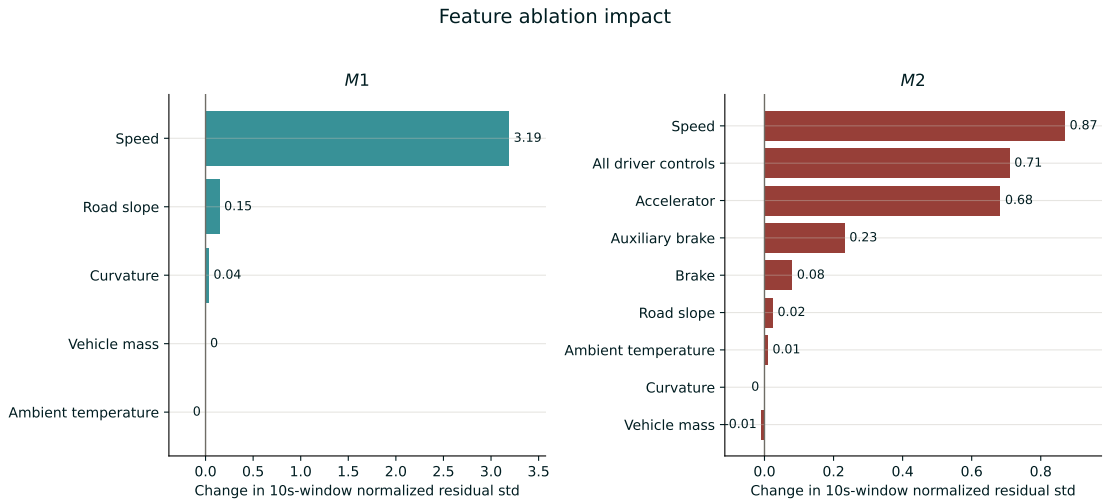


Figure 4.7: Change in weighted 10 s-window normalized residual standard deviation after feature-group ablation.

Figure 4.7 shows that speed is the dominant feature group for both model roles. Removing speed increased normalized residual spread by 3.19 for $M1$ and 0.87 for $M2$. Among the remaining $M1$ context features, road slope had the clearest effect, increasing residual spread by 0.15 when removed, while curvature, ambient temperature, and vehicle mass had only small global effects in this aggregate.

For $M2$, removing all driver-control signals increased normalized residual spread by 0.71. Among individual driver-control groups, accelerator removal had the largest effect, increasing normalized residual spread by 0.68, while auxiliary brake and brake removal increased it by 0.23 and 0.08 respectively. The available longitudinal control signals therefore improved $M2$, with accelerator features giving the strongest individual contribution. The detailed ablation and signal-availability tables are included in Appendix B.2.

4.2.5 Supervised Regression Method Suitability

The supervised regression method comparison checks whether the reported results are sensitive to method choice. It used three 17–21 t buckets: mild downhill and cold (-5 to -1 %, -10 to 0 °C), mild uphill and cold (1 to 5 %, -10 to 0 °C), and mild uphill and mild-cold (1 to 5 %,

0 to 10 °C). The comparison covers regression-based, tree-based, boosting-based, and neural-network variants, and is used to select the common supervised regression method for the reported $M1$ and $M2$ analyses. The hyperparamter settings used are listed in Table 4.4.

Table 4.4: Hyperparameter settings used in the supervised regression method comparison (same settings for $M1$ and $M2$).

Supervised regression method	**Hyperparameter settings**
XGBoost	Number of estimators: 400; learning rate: 0.05; maximum depth: 8; minimum child weight: 1.0; row subsample ratio: 0.8; column subsample ratio per tree: 0.8; L1 regularization coefficient (α): 0.0; L2 regularization coefficient (λ): 1.0
HistGradientBoosting	Number of boosting iterations: 300; learning rate: 0.05; maximum depth: 8; minimum samples per leaf: 20
Bagged trees	Number of estimators: 100; maximum depth: none; minimum samples per leaf: 1
Random forest	Number of estimators: 500; maximum depth: 12; minimum samples per leaf: 100; maximum features per split: square root of total features
Wide neural network	Hidden layer sizes: [128]; initial learning rate: 0.001; maximum training iterations: 1000
MLP (4 hidden layers)	Hidden layer sizes: [512, 256, 128, 64]; ReLU hidden activation; Adam optimizer; learning rate: 0.001; batch size: 4096; maximum training epochs: 300; early stopping
MLP (6 hidden layers)	Hidden layer sizes: [2048, 1024, 512, 256, 128, 64]; ReLU hidden activation; Adam optimizer; learning rate: 0.001; batch size: 4096; maximum training epochs: 300; early stopping
Ridge regression	Regularization coefficient (α): 1.0

Figure 4.8 summarizes the weighted 10 s-window normalized residual standard deviations for the compared regression methods. The four-hidden-layer MLP achieved 2.58 for $M1$ and 2.10 for $M2$, making it the strongest method for $M1$ on this metric, but only competitive for $M2$. The six-hidden-layer MLP did not improve this result consistently: its corresponding values were 2.62 for $M1$ and 2.06 for $M2$, indicating slightly worse $M1$ performance but a marginal improvement for $M2$. MLP being a black box model also lacks explainability when deployed. For $M2$, XGBoost still had the lowest spread (2.00), narrowly ahead of histogram gradient boosting (2.02), bagged trees (2.05), and random forest (2.06). Ridge regression was clearly weaker for both model roles, especially for normalized residual spread. Because the reported analysis uses one shared, robust, and interpretable supervised regression method for both $M1$ and $M2$, and because XGBoost is the strongest $M2$ option while remaining competitive for $M1$, the method-comparison result supports using XGBoost for the reported analyses. Detailed aggregate metrics are included in Appendix B.2, and Appendix B.3 gives the bucket-wise normalized residual comparisons for all tested supervised regression methods.

4.2.6 Union versus bucket-specific training

A model trained on pooled data from all ten primary comparable-condition buckets and the bucket-specific models are compared. For each bucket, the union model is evaluated on the same

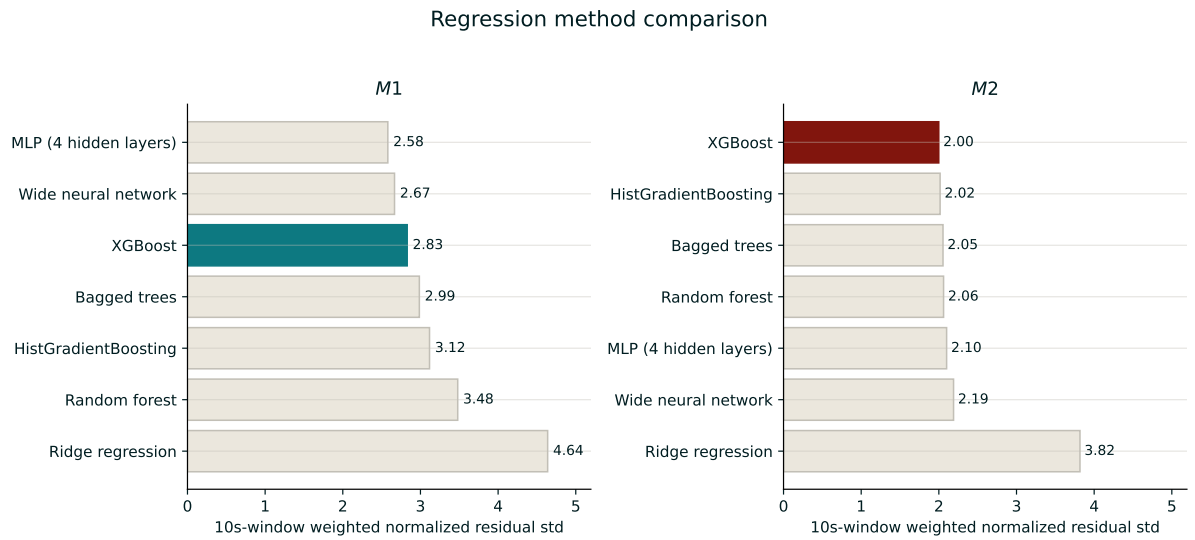


Figure 4.8: Weighted normalized residual standard deviation by supervised regression method for $M1$ and $M2$.

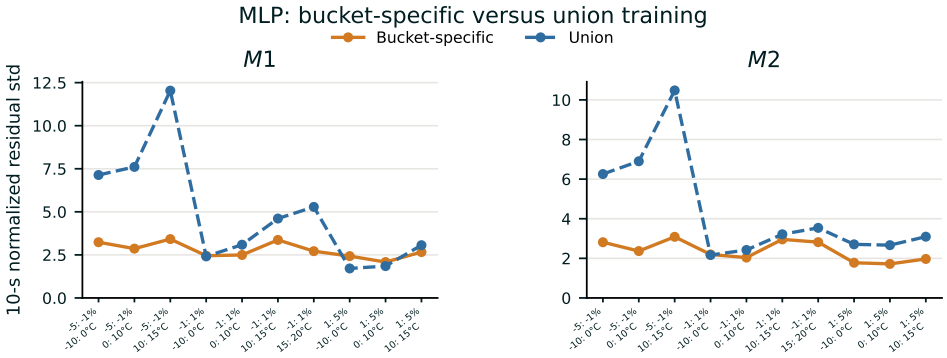


Figure 4.9: MLP comparison between bucket-specific and union training.

held-out rows used to evaluate the corresponding bucket-specific model. Thus, the comparison uses the same evaluation population and 10 s-window residual metric; only the training population changes. The union model represents pooled training across the primary buckets, whereas the bucket-specific model is trained using data from the corresponding operating-context bucket.

Figure 4.9 shows a large training-scope effect for the MLP. Bucket-specific training keeps the 10 s normalized residual standard deviation between 2.08 and 3.42 across the ten buckets, whereas union training is substantially worse in seven buckets and reaches 12.03 in the -5 to -1%, 10 to 15 °C bucket. This indicates that pooling operating contexts can substantially degrade the MLP's energy-consumption mapping. The corresponding XGBoost comparison in Figure 4.10 is much more moderate: the union and bucket-specific results remain close in many buckets, although neither training scope is uniformly better because the relative performance varies from bucket to bucket. Taken together with the regression-method comparison above, these results support XGBoost as a strong candidate regressor for energy prediction from 1 s-resolution time-series data, while retaining comparable-condition bucketing as the safer training design when operating-context effects are important.

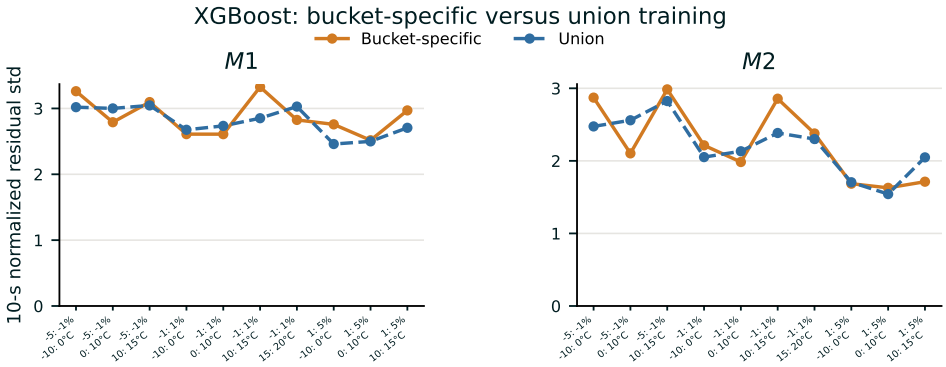


Figure 4.10: XGBoost comparison between bucket-specific and union training.

4.3 Driving-Action Model Evaluation

The final evaluation asks whether the staged energy-modeling framework can be extended from energy interpretation to driving-action prediction. This step is more challenging than the $M1$–$M2$ comparison because changing the control action also changes the simulated vehicle speed, road position, and subsequent operating context. The main evidence is therefore organized into three parts: direct $M3$ prediction quality, the contribution of preview and context features, and closed-loop simulator behavior.

4.3.1 Direct $M3$ Prediction Quality

The direct-prediction evaluation tests whether $M3$ predicts future driver-control actions more accurately than a persistence baseline. In the reported $M3$ evaluation, the target is the signed longitudinal action introduced in Section 3.5.4. This target represents propulsion and braking as one signed control variable, so the direct-prediction result is not affected by the auxiliary brake target. The target uses the accelerator/brake percentage scale, with negative values for braking and positive values for propulsion, giving an approximate range from -100 to 100 signed-control units. Mean absolute error (MAE) is used as the primary metric because it reports the average absolute distance between the predicted and recorded signed action in the same signed-control units.

The persistence baseline assumes that the future signed action is equal to the current signed action. This is a strong baseline for the 1 s-ahead target because driver controls usually change smoothly over short intervals. Figure 4.11 shows that $M3$ is therefore only marginally better than persistence at 1 s ahead, with weighted MAE values of 6.12 and 6.36 respectively. At 10 s ahead, the difference is much larger: $M3$ has a weighted MAE of 13.64, compared with 25.89 for persistence. This indicates that $M3$ learns predictive structure beyond simply holding the current control action.

The 10 s-ahead result is diagnostic, not a deployable control horizon. It is an offline point-prediction test that assumes exact replay of the recorded future context, whereas a closed-loop controller would change speed, position, and the subsequently experienced context. Its value here is therefore diagnostic: it shows that $M3$ is more informative than persistence at a longer horizon, before the feature-contribution and simulator-based evaluations test the model under stricter conditions. Detailed horizon and metric summaries are included in Appendix C.1.

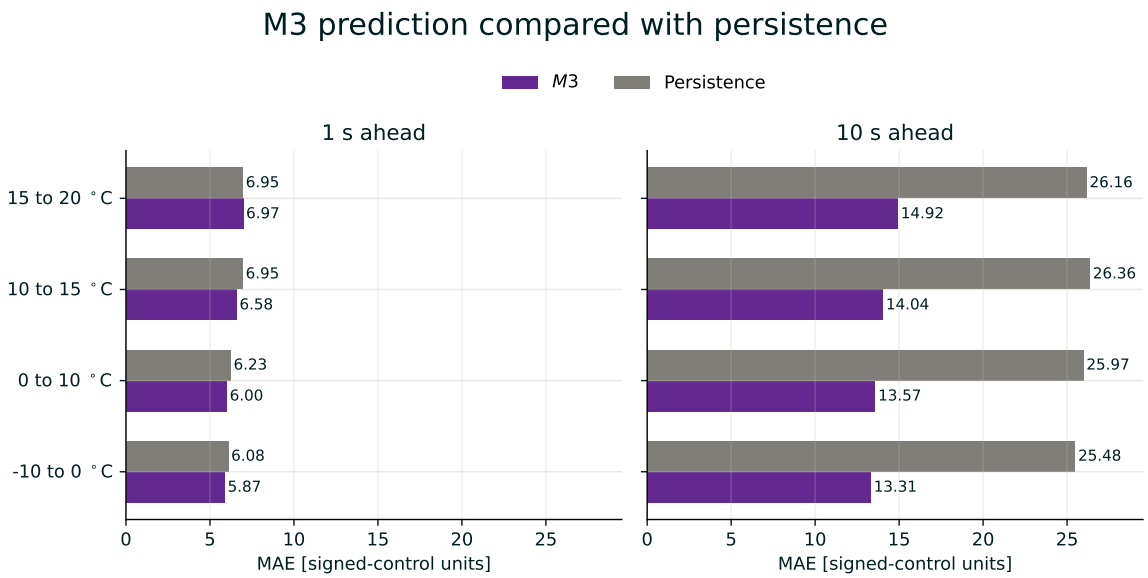


Figure 4.11: Signed-longitudinal MAE for $M3$ and a persistence baseline at 1 s and 10 s prediction horizons.

4.3.2 Preview and Context Contribution

The preview and context contribution analysis tests whether preview and context information improve $M3$ prediction compared with using only recent controls. The reported $M3$ comparison uses three variants: the full $M3$ feature set, a no-preview variant, and a recent-controls-only variant. The comparison isolates whether preview and contextual information add useful prediction signal beyond recent control history. The variant protocol follows Section 3.8.5.

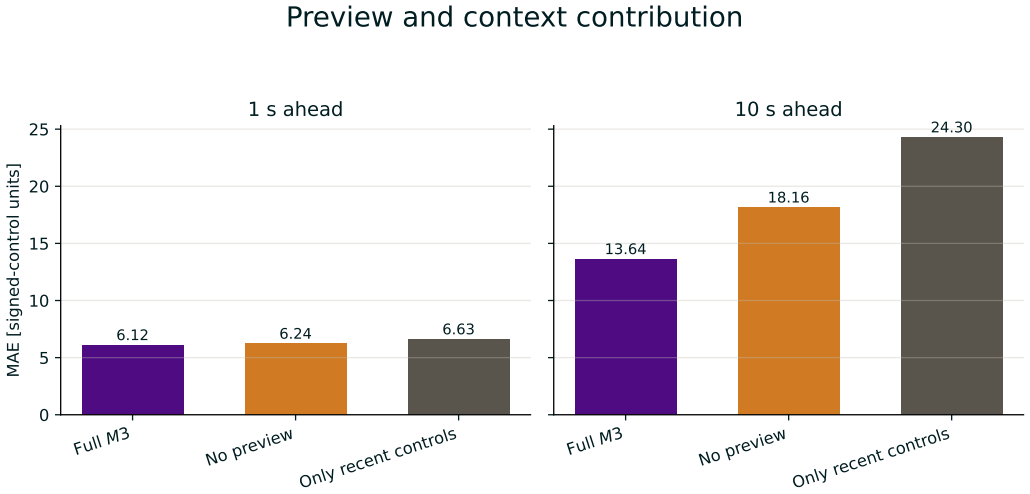


Figure 4.12: Signed-longitudinal MAE for the full $M3$ model and reduced feature variants at 1 s and 10 s ahead.

Figure 4.12 compares the same ablation variants at 1 s and 10 s ahead. At 1 s ahead, the full $M3$ model has the lowest MAE, but the differences are small: removing preview information increases MAE by 0.12 signed-control units and using only recent controls increases MAE by 0.51 units. This immediate-horizon effect is modest but consistent with the feature design: recent controls are already highly informative at 1 s ahead.

The effect becomes larger at longer diagnostic horizons. At 10 s ahead, the no-preview variant is 4.52 MAE units worse than the full model, and the recent-controls-only variant is 10.66 units worse. These results indicate that preview and contextual information add useful signal beyond recent control history, especially when the prediction target extends beyond the immediate next action. The full horizon plot and metric table are included in Appendix C.1.

The earlier $M3$-via-$M2$ counterfactual analysis is reported separately in Appendix C.2. It is useful as a diagnostic check of whether $M3$-implied controls are scored favorably by the trained control-aware energy model, but it is not treated as primary RQ3 evidence because it does not update the vehicle trajectory after changing the controls.

4.3.3 VTAB Simulator Evaluation

The $M3$-via-$M2$ counterfactual diagnostic introduced in Section 3.7.3 scores $M3$-implied controls on a fixed realized trajectory, so it cannot show how the vehicle would respond if those controls were actually followed.

Simulator-based evaluation is required because the physical consequences of changing controls cannot be fully evaluated on fixed logged rows. The simulator setups follow Sections 3.7.4.1 and 3.7.4.2. Two distinct VTAB test types are required and should not be merged. The first test reuses recorded driving data from this study by injecting or replaying the recorded operating context and comparing $M3$ against recorded-control replay and simple support controllers. The second test uses VTAB OuterWorld routes and compares $M3$ against route-aware controllers such as CCAP, which require an internal route horizon and therefore are not directly comparable on the recorded-data replay setup.

4.3.3.1 Recorded-data replay test

The recorded-data replay test asks how exploratory $M3$ controller candidates behave when logged driving windows are reproduced in VTAB as closely as possible. The replay baseline uses the recorded controls, with driver-model support where needed to maintain speed and keep the simulated vehicle aligned with the recorded distance trajectory. CCAP is not included in this test because it requires a route horizon that is not available in the same way for the replayed recorded-data setup. VTAB uses the opposite power and energy sign convention from the recorded telemetry convention used for model training, so the recorded-data EC is sign-flipped for the replay comparison before cumulative-energy traces are plotted. Energy is then compared over the post-warm-up interval (definition in Section 3.7.4.1), with cumulative energy reset to zero at the comparison start while the plots keep absolute distance on the x-axis.

Table 4.5 summarizes the training-population definitions for all $M3$ candidates considered in this section. The four residual-guided candidates are selected using either $M1$- or $M2$-based residual percentiles. The plus and minus signs refer to the residual-selection ranges (-100 to +100) and not to the sign of energy consumption. The random-25% and all-data candidates provide unstructured training-population baselines.

The replay results show that residual-based selection of $M3$ training rows produces a controllable energy–speed trade-off. The four residual-guided candidates form a visually near-linear progression in Figure 4.13, from $M3$ M1+ at 64.48 km/h and 0.50 kWh/km, through $M3$ M2+ at 68.11 km/h and 0.54 kWh/km, and $M3$ M2- at 72.11 km/h and 0.59 kWh/km, to $M3$ M1- at

Table 4.5: Training-population definitions for the $M3$ replay candidates.

Candidate	Training-population definition	Selection range or rule
$M3$ M1+	$M1$-based residual selection	+50 to +100 percentile
$M3$ M1-	$M1$-based residual selection	-100 to -50 percentile
$M3$ M2+	$M2$-based residual selection	+50 to +100 percentile
$M3$ M2-	$M2$-based residual selection	-100 to -50 percentile
Random 25%	Random sample of eligible training rows	25% of eligible rows, no residual filtering
All data	All eligible training rows	No residual filtering

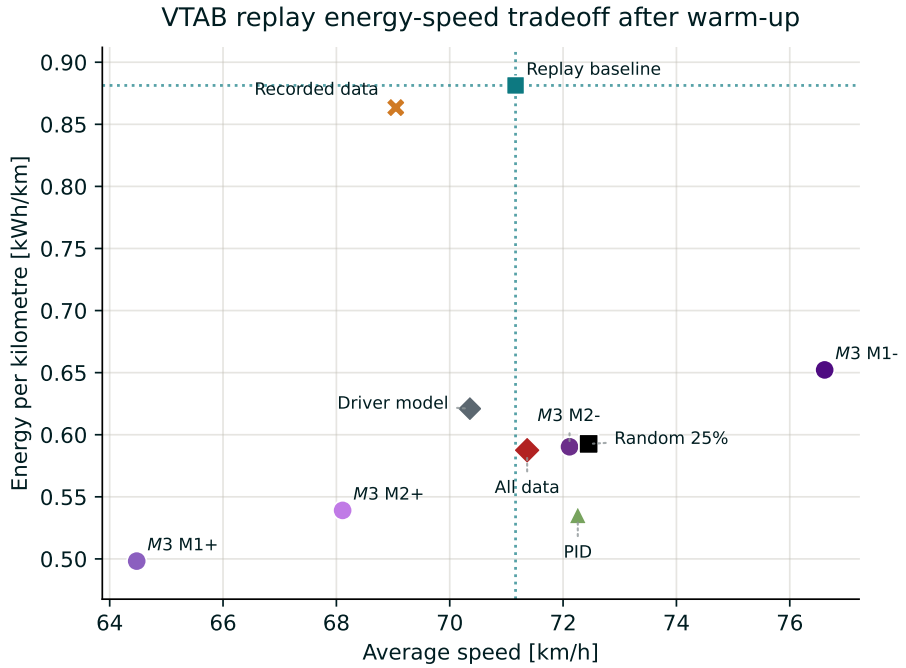


Figure 4.13: Post-warm-up energy intensity and average speed for a recorded-data replay window.

76.62 km/h and 0.65 kWh/km.

Table 4.6: Energy intensity, average speed, and percentage change relative to random 25% for a recorded-data replay window.

Candidate	**Energy [kWh/km]**	**Speed [km/h]**	**Δ energy [%]**	**Δ speed [%]**
$M3$ M1+	0.50	64.48	-15.93	-11.01
$M3$ M2+	0.54	68.11	-9.04	-5.99
PID	0.54	72.26	-9.65	-0.26
$M3$ M2-	0.59	72.11	-0.39	-0.47
All data	0.59	71.37	-0.86	-1.50
Random 25%	0.59	72.45	0.00	0.00
$M3$ M1-	0.65	76.62	10.06	5.75
Recorded human driver	0.88	71.16	48.71	-1.78

Relative to the random-25% baseline, which reaches 72.45 km/h at 0.59 kWh/km, $M3$ M1+ reduces energy intensity by 15.93% while reducing speed by 11.01%. At the other end of the progression, $M3$ M1- increases energy intensity by 10.06% while increasing speed by 5.75%. The $M3$ M2+ and M2-candidates provide intermediate points, with energy-intensity changes of -9.04% and -0.39%, respectively. This indicates that residual-based selection can shape the resulting $M3$ behavior toward either lower energy use or higher speed using the same underlying telemetry.

The random-25% and all-data baselines are close to one another: all-data reaches 71.37 km/h and 0.59 kWh/km, corresponding to only a 1.50% lower speed and a 0.86% lower energy intensity than random 25%. Their proximity indicates that unstructured sampling or using all eligible rows provides little control over the resulting $M3$ behavior. In contrast, the residual-guided candidates span a clear range of speed and energy intensity, while the PID and recorded human-driver points provide external reference behaviors.

Figure 4.13 summarizes each run by one post-warm-up point, combining its average speed and energy intensity. Located at the top of the plot, the cross represents the recorded-data reference, while the square represents the replay baseline. Figure 4.14 shows the same types of quantities as distance-resolved traces in a representative replay window: the recorded data are the black curves and the replay baseline is the blue curve, with $M3$ M1+ shown in purple. Thus, the overview provides the aggregate energy–speed comparison, whereas the trace figure shows how the corresponding speed, power, and cumulative-energy behavior develops along the route.

Figure 4.15 focuses on the same replay case between 0.7 and 1.5 km, where the route changes from uphill operation to a following downhill section. In this range, $M3$ appears to anticipate the slope transition. Both replay and $M3$ increase accelerator demand on the uphill part, but $M3$ reduces accelerator demand earlier after the crest and carries a lower, smoother speed into the downhill section. Unlike the replay baseline, which represents the recorded human-driver control pattern in this test, $M3$ avoids the brake application observed between approximately 1.28 and 1.40 km. This is relevant for energy use because regenerated braking power is not loss-free: conversion and storage losses mean that avoiding unnecessary braking can be more efficient than first adding kinetic energy and then recovering only part of it. In this hilly road segment, $M3$ is therefore observed to anticipate the downhill section and use less energy by avoiding the later braking event. An additional recorded-route replay case with full-range and zoomed trace views is provided in Appendix C.3 as supporting qualitative evidence for smoother accelerator

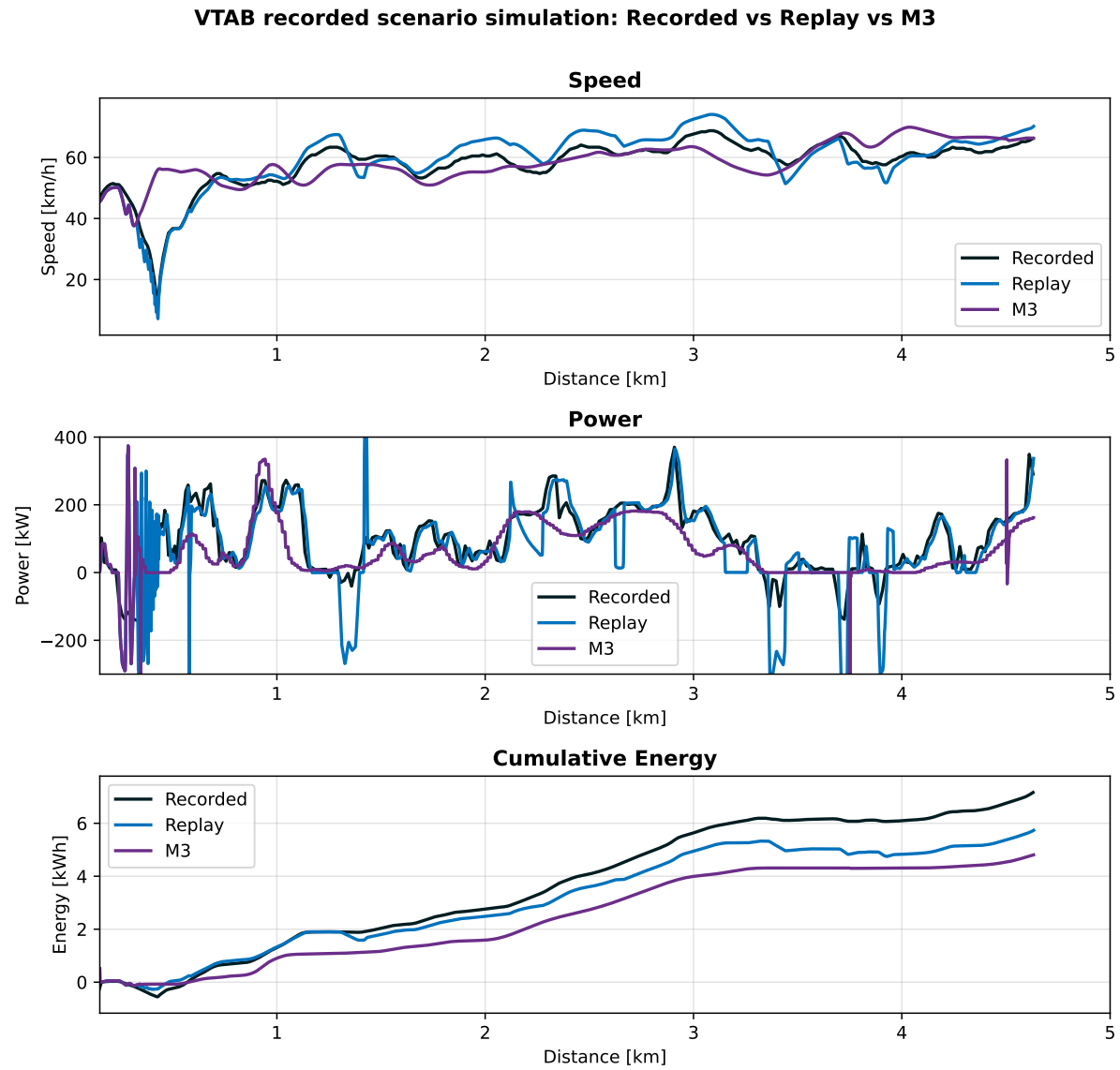


Figure 4.14: Distance-based simulation traces for recorded data, replay baseline, and $M3$.

usage and brake avoidance by $M3$.

Across the replay windows, overall terrain variability is similar (slope std $\approx 0.95\%$), yet energy outcomes still differ. This is expected because post-warm-up energy intensity depends on the combined operating context: net elevation change, slope severity and direction, ambient temperature, and control transients such as braking and acceleration variability. Therefore, no single aggregated terrain descriptor is sufficient to explain the replay-window energy differences.

4.3.3.2 OuterWorld route-controller test

The OuterWorld test asks whether $M3$ can run as an online controller on a simulator-native route that was not part of the telemetry data used to train it. The data are synthetic in the sense that the route and vehicle response come from VTAB rather than from the recorded driving dataset, but VTAB is a high-fidelity software-in-the-loop simulator as described in Section 3.7.4. The test is therefore used as a new-condition closed-loop check, not as final vehicle-validation evidence. It is also the appropriate comparison point for CCAP because the OuterWorld route exposes the horizon information that CCAP needs internally. As in the replay test, the first 30 s are excluded as warm-up and energy is compared over the common post-warm-up distance interval.

Table 4.7: OuterWorld route-controller post-warm-up cumulative-energy summary.

Controller	Dist. [km]	Energy [kWh]	Delta vs CCAP [kWh]	Avg. speed [km/h]
Driver model	49.80	51.17	3.94	89.47
CCAP	49.80	47.24	0.00	89.07
PID	49.80	52.17	4.93	93.53
$M3$	49.80	50.37	3.13	89.18

The OuterWorld result shows both the useful and limited sides of $M3$. Figure 4.16 reports the cumulative-energy difference relative to $M3$, so negative values indicate lower energy than $M3$ and positive values indicate higher energy. The useful result is that $M3$ can independently drive the simulated truck on a new route: it completes the route and uses 50.37 kWh over the post-warm-up interval, which is 1.80 kWh lower than PID and 0.81 kWh lower than the driver-model baseline. However, it does not outperform CCAP, which uses 3.13 kWh less than $M3$. Figure 4.17 shows the main limitation: $M3$ does not explicitly follow the route speed profile followed by CCAP. It is slower than CCAP around 15–21 km and again near 24–26 km, but it also overspeeds strongly around 28–32 km and more briefly around 47–49 km. The OuterWorld test therefore supports $M3$ as a plausible learned driving-action controller in simulation, but not as a deployable speed-compliant controller.

Boosted-tree action models' partition-based predictions typically become constant beyond observed feature values rather than following the underlying physical trend [45]. The speed range in the training data evidently are not well above the local European truck speed limit of 90 km/h. The $M3$ driving-action model's resulting speed in especially continuous downhill sections moves the simulated state outside the training support.

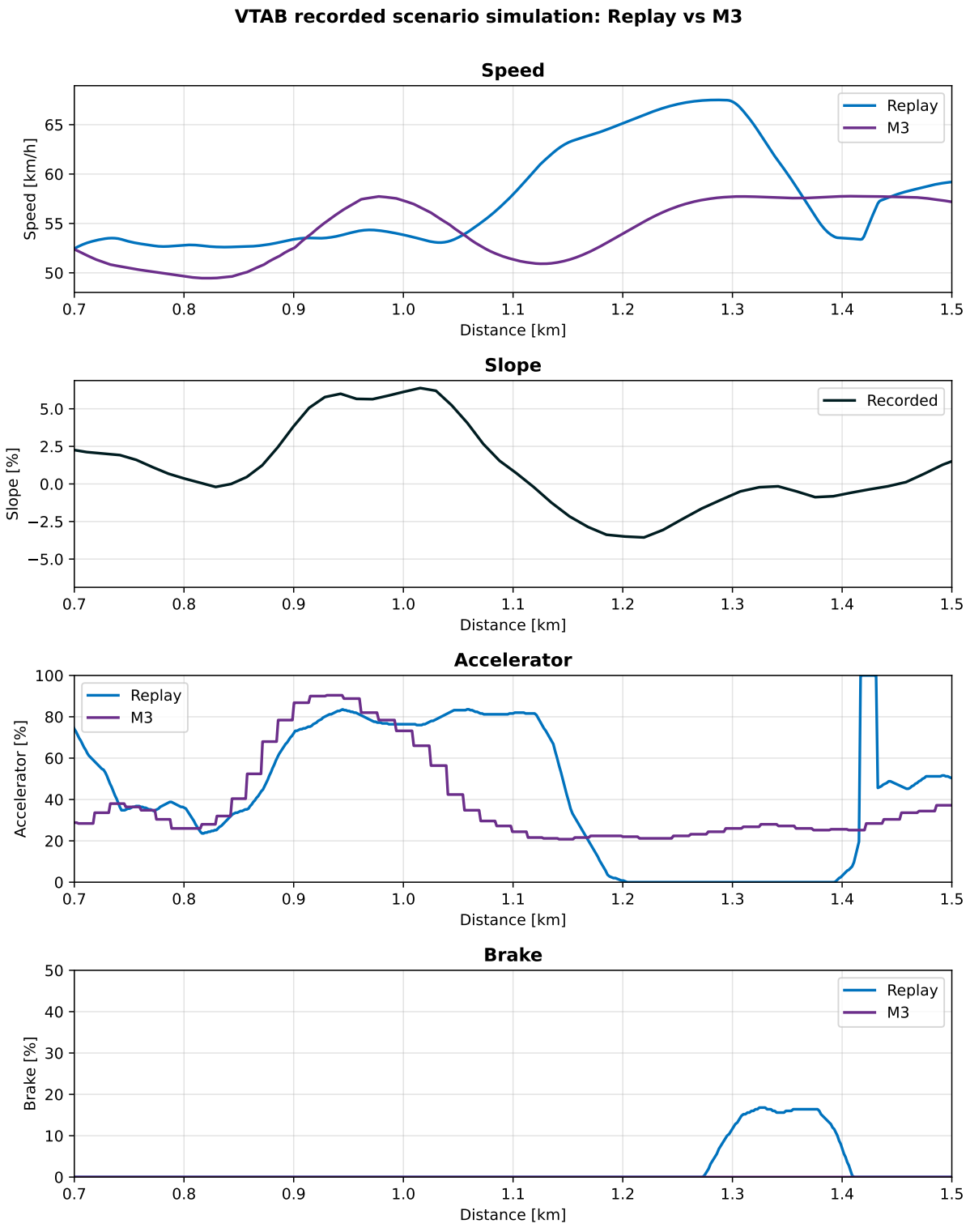


Figure 4.15: Zoomed simulation traces from 0.7 to 1.5 km.

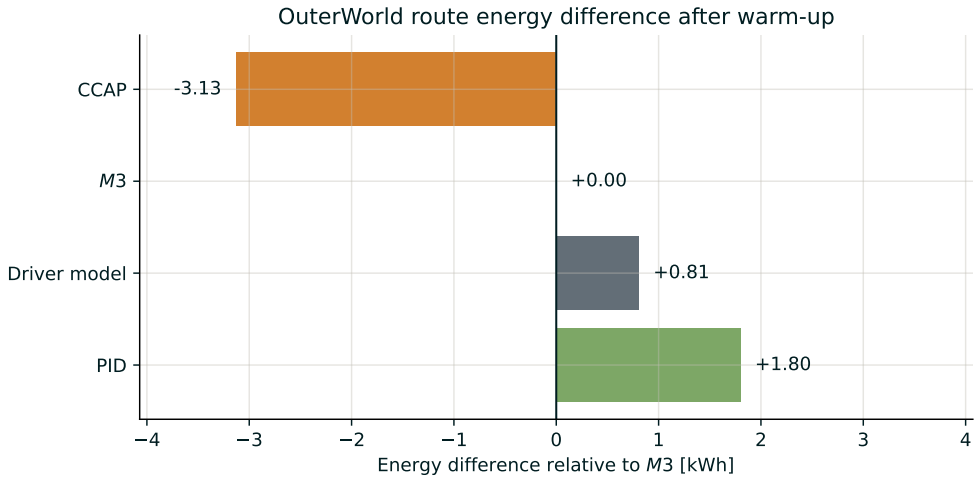


Figure 4.16: Post-warm-up cumulative-energy difference on the VTAB OuterWorld route.

4.3.4 Driving-Action Summary

For direct action prediction, Section 4.3.1 shows that $M3$ is clearly better than persistence for the 10 s-ahead and next-10 s-mean signed-longitudinal targets, but only marginally better for the immediate 1 s-ahead target. The driving-action model therefore improves on persistence for longer support horizons, while the immediate action prediction remains close to a strong short-horizon persistence baseline.

For feature contribution, Section 4.3.2 shows that the full $M3$ model had lower MAE than both the no-preview and recent-controls-only variants, with the gap increasing at longer horizons. This indicates that preview and contextual information improve signed-longitudinal action prediction beyond recent controls alone.

For closed-loop behavior, Section 4.3.3 gives mixed but useful simulator evidence. In recorded-data replay, exploratory $M3$ candidates reduced energy intensity relative to the replay baseline, and the zoomed hilly segment shows a plausible mechanism: $M3$ anticipates the following downhill section and avoids the braking event seen in replay. On the OuterWorld route, $M3$ also completes a new simulator-native route and consumes less energy than the PID and driver-model baselines, but it uses more energy than CCAP and does not explicitly follow the route speed profile. Overall, RQ3 is answered positively for prediction and feature-contribution evidence, and partially for simulator evidence: $M3$ can produce plausible learned driving actions in simulation, but the current XGBoost controller is not a deployable speed-compliant controller. Explicit reference tracking and constraint handling on $M3$ could make it closer to deployment.

4.4 Discussion

The energy-modeling results support the main premise of the thesis: driver-related contribution to energy consumption can be estimated more defensibly when operating context is modeled first. The matched $M1$–$M2$ comparison reduced both RMSE and normalized residual spread in every primary 17–21 t comparable-condition bucket. Because $M1$ and $M2$ use the same target, the same evaluation rows, and the same bucket definitions, the improvement from $M2$ supports the interpretation that the available driver-control signals add explanatory value beyond the selected

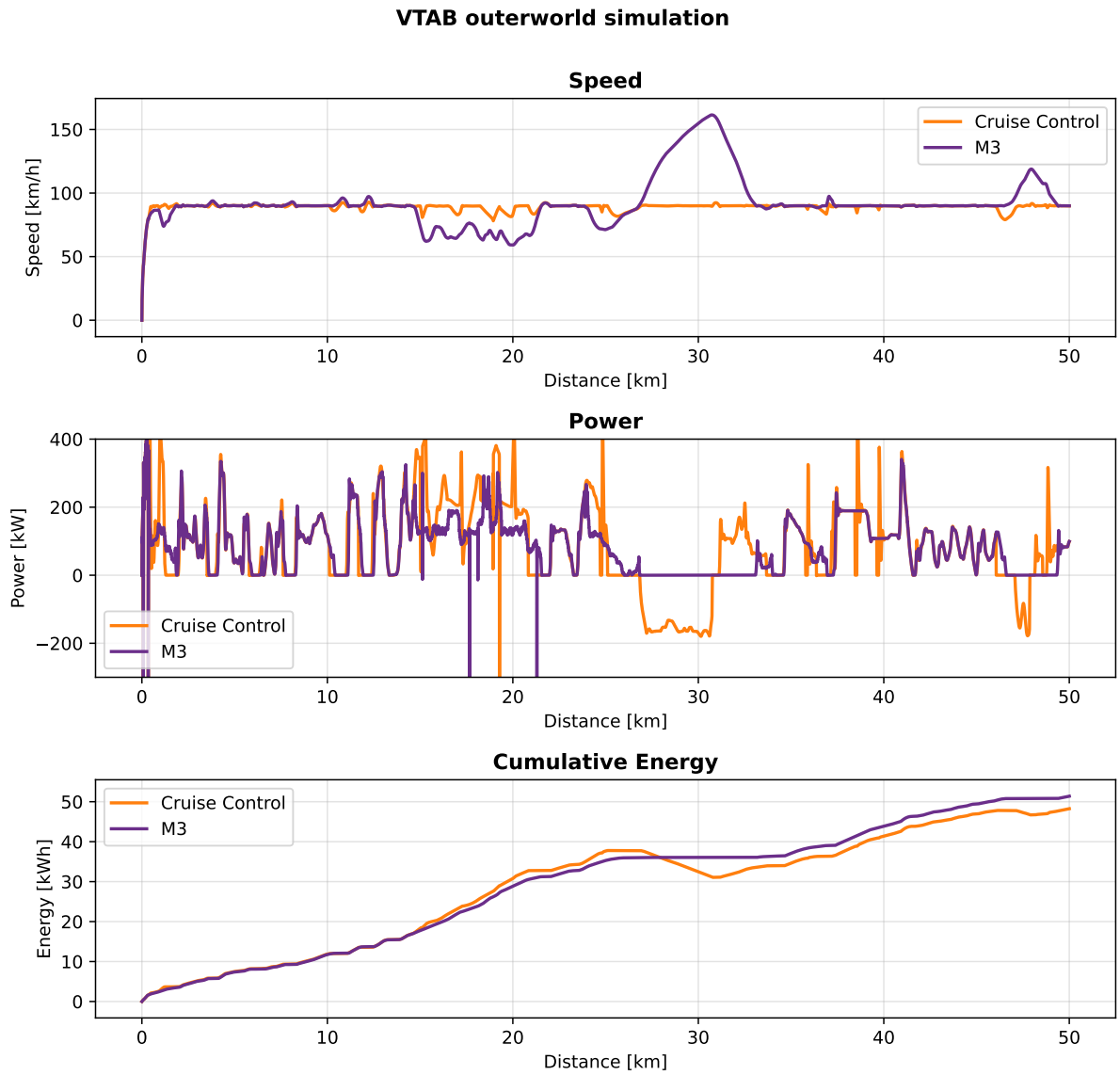


Figure 4.17: Distance-based OuterWorld simulation traces for CCAP and $M3$.

context variables. Within this scope, RQ1 is therefore supported. The result should still be read as predictive and explanatory evidence, not as a causal estimate of how much energy a different driver action would have saved.

The operating-scenario results show why the comparison should not be collapsed into a single global driver score. Residual spread and $M1$–$M2$ improvement varied across slope and temperature ranges. The strongest normalized-spread reduction appeared in colder mild-uphill operation, while the weakest appeared in the warmest flat bucket. This pattern is technically plausible because uphill energy use is more directly tied to propulsion demand and accelerator behavior, while downhill and flat cases can involve more coasting, braking, route constraints, and unobserved states. The practical implication is that driver-related interpretation should remain tied to comparable operating conditions.

The robustness and data-sensitivity results answer RQ2 as a set of practical requirements for the tested framework. The final bucket design should control road slope, ambient temperature, and vehicle mass while preserving enough data for grouped training and evaluation. Feature resolution of 1–5 s was sufficient in the reported tests; at 10 s the context-only model degraded more clearly, so 10 s should be treated as a coarser and less desirable boundary rather than the preferred resolution. The training-volume test suggests that roughly 40 comparable hours per bucket before splitting, corresponding to about 24 sampled training hours under the grouped split used here, is a useful lower guideline for similar bucket-wise training. This is not a saturation point, because additional data can still improve some cases.

The feature-ablation results define the minimum signal scope more concretely. Vehicle speed is a mixed variable because it reflects driver behavior, route constraints, traffic, and vehicle dynamics. Retaining speed in both models is nevertheless important for a direct comparison, because it keeps the main operating-state information constant while isolating the additional contribution of the driver-control signals introduced in $M2$. Speed was the most influential feature group for both model roles, although its effect was substantially smaller for $M2$. Road slope was the strongest remaining context feature in the aggregate $M1$ ablation. For $M2$, removing all driver-control signals worsened the result, and accelerator pedal features were the strongest individual driver-control group for energy-consumption prediction. Brake pedal and auxiliary brake lever features also contributed, but less strongly in the aggregate ablation. XGBoost was retained as a pragmatic common supervised regression method because it gave the strongest $M2$ result and remained competitive for $M1$.

The supervised regression method decision also followed a deployment-aware criterion, not only an accuracy ranking. The tested alternatives included a wide neural-network baseline, but tree-ensemble methods provided a better combined tradeoff across predictive performance, training stability, and reproducible bucket-wise evaluation under the current tabular feature design. This criterion is relevant because the intended application direction is onboard driver support, where bounded compute and memory budgets, deterministic inference behavior, and integration simplicity are practical constraints. Within this scope, retaining XGBoost as the shared $M1$/$M2$ method is therefore a methodological choice aligned with both the empirical comparison and the expected deployment setting.

The driving-action results answer RQ3 more cautiously. The direct-prediction and ablation results show that $M3$ learns useful control-action structure, while the simulator evaluation shows how the training-data selection rule can shape the behavior of the resulting candidates. Selecting training rows according to positive or negative residual ranges from energy consumption prediction models produces candidates with different energy–speed profiles. In replay, the four

residual-guided candidates progress from lower-energy and slower behavior to higher-speed and more energy-intensive behavior. This demonstrates that the method can generate a family of driving-action candidates rather than only one fixed $M3$ behavior.

The random-25% and all-data baselines provide an important contrast. Their results are very close, indicating that random sampling and unfiltered training provide little deliberate control over the resulting behavior. The separation of the residual-guided candidates therefore comes from the selection criterion rather than simply from changing the training-population size. The method should consequently be understood as an outcome-guided candidate-generation framework: the energy model and residual range used for selection influence the driving priority expressed by the trained $M3$ candidate.

This also motivates alternative objective models. Future studies could replace the current energy-residual criteria with objectives that represent different priorities, such as speed preservation, comfort, braking avoidance, emissions, or a weighted combination of energy and performance. Such objectives could drive the same selection procedure toward candidates with different behavioral preferences. In the OuterWorld test, $M3$ can independently complete a new simulator-native route, but it does not explicitly follow the route speed profile and therefore cannot be interpreted as a deployable controller. Thus, driving-action results are best read as evidence that the staged framework can generate plausible driving-action candidates, not as evidence that the current XGBoost controller is ready for speed-compliant driver support.

These conclusions are bounded by the available data and model scope. The comparable-condition buckets reduce major context confounding, but they do not eliminate unmeasured traffic, route, vehicle-configuration, or driver-state effects. Further splitting the cohort by configuration details such as roof height, axle ratio, gearbox type, or application could reduce residual vehicle variation, but it would also reduce the available data in each comparable-condition bucket. Given the training-volume results, this tradeoff is part of the validity of the reported design: the cohort was kept practically comparable while preserving enough data for grouped training and evaluation. Steering angle was unavailable, so lateral driver-control behavior is outside the reported control-aware model. The simulator evidence is also bounded: VTAB provides a stronger closed-loop test than offline prediction. The framework is therefore validated as a context-aware telemetry and simulator-evaluation method for the selected truck cohort, bucket design, signal set, and evaluation protocol, rather than as a universal causal driver-efficiency measure or deployable driving controller.

Chapter 5

Conclusions and Future Work

5.1 Conclusions

The purpose of this thesis was met at research-prototype level. The work implemented comparable-condition buckets, a context-only energy model ($M1$), a control-aware energy model ($M2$), and a driving-action model ($M3$). Together, these stages separate operating context from available driver-control signals before testing whether the resulting energy-relevant information can support future control-action prediction.

RQ1 is supported within the available telemetry and evaluation design. Adding driver-control signals improved prediction compared with the context-only model in all primary comparable-condition buckets. This shows that the available driver-control signals contain predictive information beyond the selected operating-context variables. The result is not a causal estimate of individual driver actions, but it is more defensible than comparing raw energy consumption across trips or drivers.

RQ2 is supported as practical guidance rather than as universal thresholds. Reliable context-aware energy modeling required comparable operating conditions, sufficient data per bucket, and feature groups that represent both operating context and driver-control behavior. In this study, slope–temperature–mass buckets were the most useful bucket family, 1–5 s feature resolution was preferable to coarser inputs, and roughly 40 comparable hours per bucket before splitting was a useful lower guideline. Speed and road slope were the most important context-related feature groups, while accelerator pedal features were the strongest individual driver-control group.

RQ3 is partially supported. $M3$ predicted longer-horizon actions more accurately than persistence and benefited from preview and context information. In replay simulation, it showed plausible energy-saving behavior by reducing accelerator demand earlier and avoiding braking before a downhill section. It independently completed a new simulator-native route and used less energy than the PID and driver-model baselines, but it did not outperform Cruise Control with Active Prediction or explicitly follow the route speed profile. The present $M3$ is therefore a promising driving-action candidate generator, not a deployable speed-compliant controller.

The staged framework allows training different speed - energy profiles of driving-action models through residual based selection of training data.

5.2 Limitations

The $M1$–$M2$ evidence is predictive and explanatory, not causal. Lower prediction error for $M2$ shows that driver-control signals add information beyond the selected operating-context variables, but it does not prove how much energy would have changed under a different action. Comparable-condition buckets reduce major context differences, but they do not remove all confounding.

The signal scope also limits the interpretation. Steering angle, detailed traffic interaction, richer vehicle-configuration details, and driver-state information were not included. Speed is especially important because it is a mixed variable: it reflects route, traffic, vehicle dynamics, and driver behavior at the same time. It improves prediction, but it cannot be treated as purely context or purely driver control.

There is also a tradeoff between comparability and data volume. More detailed bucket definitions could separate operating conditions more precisely, but each split reduces the data available for training and evaluation. The selected slope–temperature–mass design is therefore a practical balance for this dataset, not a universally optimal bucket definition.

Finally, the $M3$ evidence is simulator-bound. VTAB closed-loop simulation is stronger than offline scoring because the vehicle trajectory evolves from the controller's own actions, but it is still synthetic and does not replace vehicle validation. The present XGBoost controller predicts an immediate longitudinal action and does not enforce route constraints, speed limits, comfort constraints, or safety rules. Overall, the results apply to the selected truck cohort, signal set, bucket design, model definitions, grouped evaluation protocol, and VTAB scenarios, not to a universal driver-efficiency measure or production controller.

5.3 Future Work

The feature set should first be extended with steering and lateral-control signals, and more detailed vehicle-configuration variables.

A separately specified context-only energy model could be developed that excludes vehicle speed while enriching the remaining operating-context representation. Unlike the existing speed-ablation test, this model would add road-related variables such as speed limits, stop signs, intersections, road type, and road quality. These features could help describe situations in which braking is expected from the road layout or regulations, providing a stronger basis for interpreting whether residual driver-control effects reflect discretionary braking or omitted road demands. Comparing this richer context-only model with $M2$ could provide a more informative assessment of the additional explanatory value of the observed driver-control signals. Implementing this extension would be challenging because deriving such road-context variables requires vehicle-location data, which is sensitive and subject to privacy and data-governance restrictions; sufficiently broad operating-context coverage would therefore require careful data protection, aggregation, and approval procedures.

Future work should then make $M3$ more controller-aware. A practical next step is to wrap the immediate-action model with constrained logic for speed-limit enforcement, acceleration and braking smoothness, route-horizon optimization, and safety guardrails. Reinforcement learning could also reward lower energy consumption while penalizing under-speeding, over-speeding,

excessive braking, and uncomfortable control changes.

The simulator evaluation should then be expanded to more VTAB routes, slopes, temperatures, masses, and traffic-like situations. If the evidence remains promising, hardware-in-the-loop or controlled vehicle testing should follow under clear safety supervision. Because the intended application direction is onboard driver support with practical runtime constraints, future supervised-regression-method extensions should be evaluated under explicit latency and memory budgets, then compared against the current XGBoost baseline using the same grouped protocol.

Finally, hybrid machine-learning, physics, and controller approaches should be investigated as alternatives to immediate-action XGBoost.

5.4 Reflections

The broader relevance of this work is environmental, economic, and ethical. Better energy-aware driving can reduce energy demand and support more efficient electrified freight transport. Lower energy use may also reduce operating cost, and smoother longitudinal control may reduce stress on brakes, tyres, and drivetrain parts, but these effects were not measured directly.

The main ethical issue is fairness. Raw energy consumption is not a fair driver score because it is strongly affected by route, slope, temperature, mass, traffic, and vehicle configuration. Driver feedback should therefore be context-aware, transparent about uncertainty, and designed to support drivers rather than blame or rank them without accounting for operating context. The driving-action model should also be understood as driver support, not as replacement of driver authority or as autonomous driving.

This work is most closely related to UN sustainability goals SDG 9 and SDG 13 on industry, innovation, infrastructure, and climate impact. Its contribution is modest but concrete: it develops a method for using telemetry more responsibly in energy-aware truck operation while emphasizing that such methods must be transparent, context-aware, and carefully validated before real driver-support use.

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Appendix A

Energy-Influence Supporting Results

A.1 Detailed Energy-Influence Diagnostics

This section collects detailed support for the driver-related energy-influence result: the selected-bucket residual distribution, evaluation-scale robustness, grouped-split robustness, and the higher-mass sanity check. All results in this section are produced with the final XGBoost-based $M1$ and $M2$ models.

Figure A.1 gives a distribution-level view of the selected mild-uphill, cold bucket (1 to 5 %, -10 to 0 °C). Higher empirical cumulative distribution function (ECDF) curves indicate that a larger share of rows falls below a given residual threshold. The figure confirms that $M2$ shifts more of the held-out rows toward lower absolute normalized residuals, consistent with the actual-vs-predicted view in the main energy-influence section.

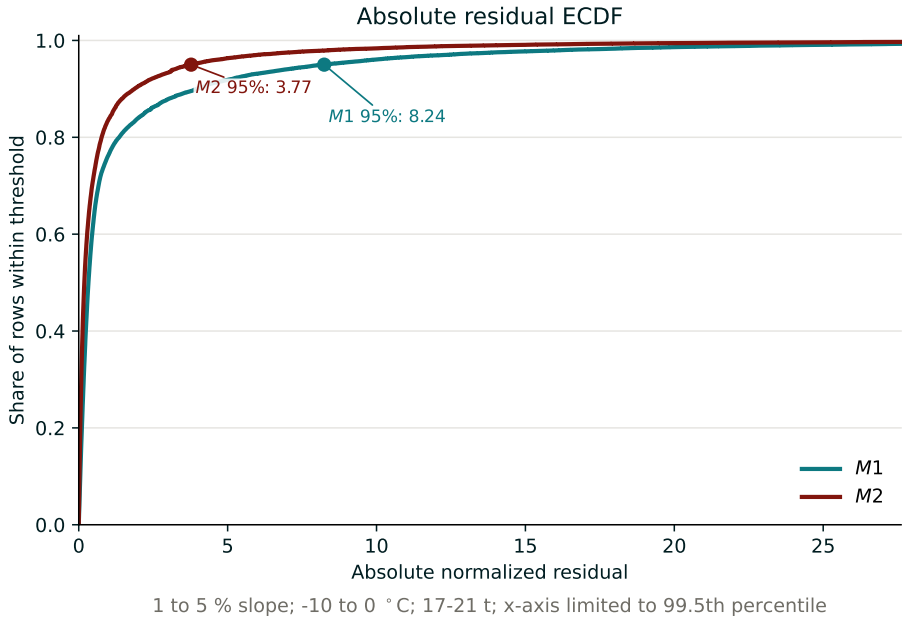


Figure A.1: Absolute normalized residual ECDF for the selected 17–21 t bucket.

Table A.1 gives the row-level and window-level residual-spread changes by bucket. The negative deltas show that the $M2$ advantage remains directionally stable across evaluation scales, while

the magnitude varies by bucket and window length.

Table A.1: Change in normalized residual standard deviation after adding driver-control variables at row-level and window-averaged evaluation scales. Values are $M2$ minus $M1$.

Slope (%)	Temperature (°C)	Row	10 s	20 s	60 s
-5 to -1	-10 to 0	-0.41	-0.64	-0.77	-0.65
-5 to -1	0 to 10	-0.31	-0.62	-0.55	-0.56
-5 to -1	10 to 15	-0.58	-0.55	-0.43	-0.83
-1 to 1	-10 to 0	-0.69	-0.68	-0.71	-0.50
-1 to 1	0 to 10	-1.19	-0.44	-0.52	-0.80
-1 to 1	15 to 20	-0.87	-0.19	-0.71	-0.59
-1 to 1	10 to 15	-1.88	-0.51	-0.43	-0.79
1 to 5	-10 to 0	-1.21	-1.11	-1.57	-2.09
1 to 5	0 to 10	-2.04	-1.04	-0.61	-0.55
1 to 5	10 to 15	-2.89	-0.44	-1.19	-1.61

Figure A.2 and Table A.2 give the grouped-split robustness details. The repeated negative deltas show that the result is not caused by one favorable grouped split.

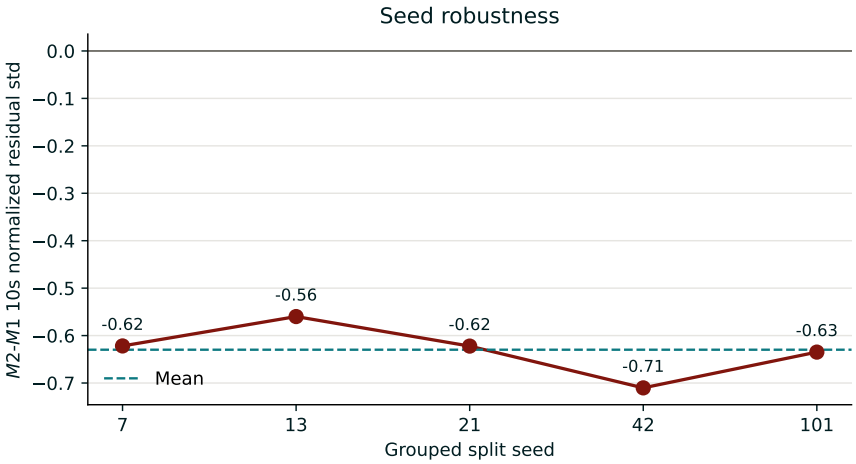


Figure A.2: Grouped-split seed robustness for normalized residual spread.

Table A.3 and Figure A.3 give the higher-mass sanity check. In the figure, horizontal lines show the bucket-level minimum-to-maximum range and points show the mean across buckets. The available 35–39 t buckets show the same direction of $M1$–$M2$ improvement, although fewer comparable-condition buckets are available than in the primary 17–21 t range.

Table A.2: Weighted $M1$–$M2$ comparison across grouped split seeds for the primary bucket set. Normalized residual columns use 10 s windows; Δ columns are $M2$ minus $M1$.

Seed	$M1$ row RMSE	$M2$ row RMSE	Δ row RMSE	$M1$ 10 s norm. res. std	$M2$ 10 s norm. res. std	Δ 10 s norm. res. std
7.00	53.74	40.91	-12.83	2.92	2.30	-0.62
13.00	52.92	40.31	-12.61	2.81	2.25	-0.56
21.00	53.11	40.47	-12.64	2.93	2.31	-0.62
42.00	53.65	40.43	-13.22	2.78	2.07	-0.71
101.00	53.95	41.37	-12.58	2.88	2.24	-0.63

Table A.3: 10 s-window normalized residual summary for the available 35–39 t comparable-condition buckets. Δ is $M2$ minus $M1$.

Slope (%)	Temperature (°C)	Data hours	$M1$ 10 s mean	$M1$ 10 s std	$M2$ 10 s mean	$M2$ 10 s std	Δ 10 s std
-5 to -1	-10 to 0	40	-0.96	5.37	-0.49	4.59	-0.78
-5 to -1	0 to 10	42	-1.17	6.55	-0.54	4.26	-2.30
-1 to 1	-10 to 0	74	0.66	3.34	0.18	2.12	-1.22
-1 to 1	0 to 10	85	0.52	4.06	0.08	2.87	-1.19
1 to 5	0 to 10	42	0.48	3.32	0.18	2.23	-1.09

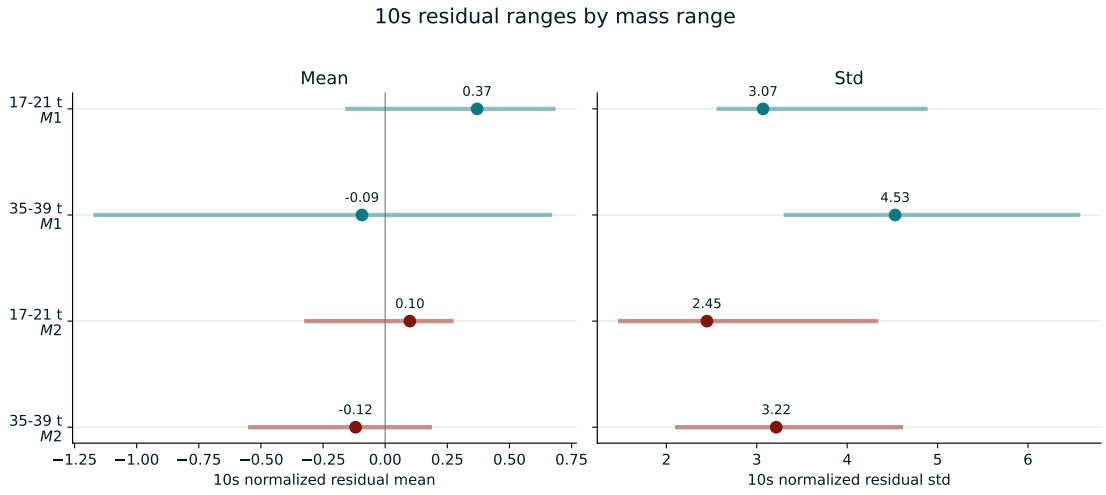


Figure A.3: Normalized residual ranges by mass range.

Appendix B

Data-Requirements Supporting Results

B.1 Bucket-Definition Sensitivity Details

The main results chapter reports the bucket-definition sensitivity study through the family-selection and range-refinement figures. Table B.1 gives the controlled family-selection values from the bucket-combination test. Table B.2 gives the later range-refinement values used to choose the final slope, temperature, and mass ranges. The values are weighted by available bucket rows.

Table B.1: Controlled bucket-family selection results.

Bucket family	Buckets	Available h	$M1$ RMSE	$M2$ RMSE	$M1$ norm. res. std	$M2$ norm. res. std
Slope only	2	1704	29.96	26.47	3.02	2.69
Temperature only	5	3955	31.64	24.84	3.02	2.56
Mass only	2	2058	29.60	21.83	3.22	2.57
Slope + temperature	12	3679	31.39	24.76	2.96	2.50
Slope + mass	6	1961	28.74	20.57	3.08	2.32
Temperature + mass	7	1929	26.26	21.01	3.18	2.79
Slope + temperature + mass	9	1065	21.62	18.72	2.38	2.12

Table B.2: Bucket-range refinement results.

Refinement	Variant	Buckets	Available h	$M1$ RMSE	$M2$ RMSE	$M1$ delta	$M2$ delta
Slope bins	Coarse bins	9	1065	21.62	18.72	0.00	0.00
Slope bins	Finer bins	15	1015	22.07	19.21	0.44	0.49
Temperature bins	10-20 C	1	135	17.81	15.88	0.78	1.26
Temperature bins	10-15 / 15-20 C	2	135	17.03	14.61	0.00	0.00
Mass width	18-20 t	4	300	21.55	18.89	0.97	1.16
Mass width	17-21 t	4	514	20.59	17.74	0.00	0.00
Mass width	16-22 t	4	616	21.36	18.44	0.78	0.70
Mass width	15-24 t	4	709	21.62	18.90	1.03	1.16

B.2 Detailed Data-Requirements Diagnostics

This section collects detailed numeric support for the data-requirements results reported in the main text. Unless explicitly identified as part of the supervised-regression-method comparison, the results in this section use the final XGBoost-based $M1$ and $M2$ models.

Figure B.1 and Table B.3 give the 60 s-window temporal-resolution check.

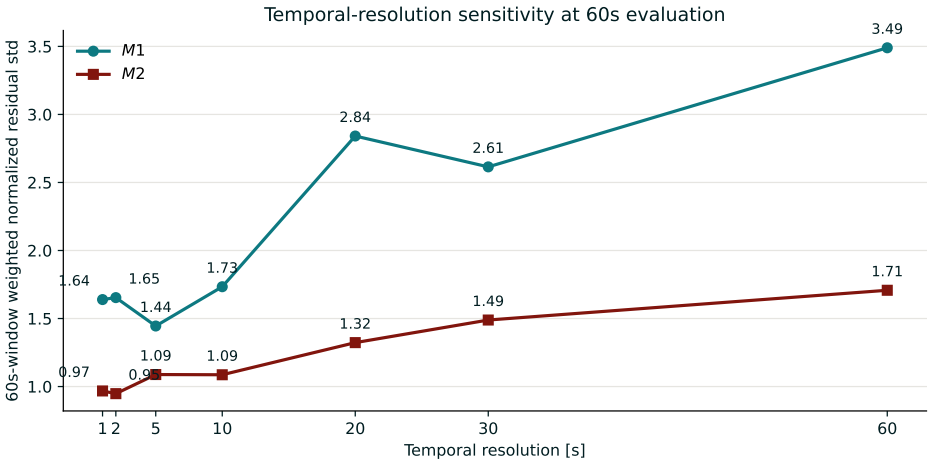


Figure B.1: Weighted 60 s-window normalized residual standard deviation for $M1$ and $M2$ under feature resolutions from 1 s to 60 s.

Table B.3: Extended temporal-resolution sensitivity evaluated with 60 s window-level averages. Normalized residual columns use 60 s windows; Δ is $M2$ minus $M1$.

Resolution (s)	$M1$ **60 s norm. res. std**	$M2$ **60 s norm. res. std**	Δ **60 s norm. res. std**
1.00	1.64	0.97	-0.67
2.00	1.65	0.95	-0.71
5.00	1.44	1.09	-0.36
10.00	1.73	1.09	-0.65
20.00	2.84	1.32	-1.52
30.00	2.61	1.49	-1.13
60.00	3.49	1.71	-1.78

The 60 s-window values reinforce the same practical boundary as the main 10 s result: moderate coarsening is acceptable, but very coarse feature updates lose useful temporal information, especially for the context-only model.

Table B.4 gives the numeric feature-ablation deltas behind Figure 4.7.

Table B.5 defines the signal scope used to interpret the ablation result. It clarifies that the control-aware model is a model of the available longitudinal and auxiliary driver-control signals, not a complete model of all possible driver actions.

Together, the ablation and availability tables show both sides of the feature requirement: the available driver-control signals add useful information, but the model scope remains limited by missing lateral-control and traffic-interaction signals.

Table B.6 gives the aggregate supervised-regression-method metrics behind Figure 4.8.

The supervised-regression-method table supports the XGBoost choice mainly through the $M2$ result, while also showing why the choice should be treated as pragmatic rather than universally optimal for every model role.

The later MLP comparison extends this model-family check across the ten primary comparable-condition buckets. It evaluates the smaller and larger feed-forward MLP configurations using

Table B.4: Selected 10 s-window feature-ablation results for the context-only and control-aware models. Positive changes indicate larger residual spread than the corresponding full model.

Model	Removed feature group	10 s norm. res. std	Δ 10 s norm. res. std
$M1$: context-only	Speed	5.97	3.19
$M1$: context-only	Road slope	2.93	0.15
$M1$: context-only	Curvature	2.82	0.04
$M1$: context-only	Ambient temperature	2.78	-0.00
$M1$: context-only	Vehicle mass	2.78	0.00
$M2$: control-aware	Speed	2.94	0.87
$M2$: control-aware	All driver controls	2.78	0.71
$M2$: control-aware	Accelerator	2.75	0.68
$M2$: control-aware	Auxiliary brake	2.30	0.23
$M2$: control-aware	Brake	2.15	0.08
$M2$: control-aware	Road slope	2.10	0.02
$M2$: control-aware	Ambient temperature	2.08	0.01
$M2$: control-aware	Vehicle mass	2.06	-0.01
$M2$: control-aware	Curvature	2.07	-0.00

Table B.5: Availability and modeling role of signal groups used in the analysis.

Signal group	Status	Thesis interpretation
Road slope, ambient temperature, vehicle mass	Available and used	Main operating-context variables used for comparable-condition bucketing and context-aware modeling.
Speed and curvature	Available or derived and used	Operational variables that improve prediction, but are not interpreted as purely driver-controlled because they also reflect route, traffic, and vehicle dynamics.
Accelerator, brake, auxiliary brake lever	Available and used	Main longitudinal and auxiliary driver-control signals used to test whether control-aware modeling improves energy-consumption prediction.
Steering angle	Unavailable	Important lateral driver-control variable; its absence limits the completeness of the driver-control representation.
Detailed traffic interaction	Not used in the reported energy models	Potentially relevant operating context, but not included in the final $M1$–$M2$ energy-model comparison because the available feature set did not provide a sufficiently defensible representation for that analysis.

Table B.6: Weighted regression method comparison for $M1$ and $M2$. 10 s-window normalized residual standard deviation is the primary comparison metric; – indicates that the corresponding row-level RMSE was not retained for the MLP summary.

Model family	$M1$ **10 s** norm. res. std	$M1$ row RMSE	$M2$ **10 s** norm. res. std	$M2$ row RMSE
XGBoost	2.83	58.12	2.00	44.43
HistGradientBoosting	3.12	61.60	2.02	44.44
Bagged trees	2.99	61.75	2.05	45.06
Random forest	3.48	66.57	2.06	45.75
Wide neural network	2.67	55.41	2.19	45.08
MLP	2.58	55.43	2.10	44.81
Ridge regression	4.64	62.46	3.82	51.24

both bucket-specific and all-bucket training populations, while keeping the feature and target contracts unchanged.

B.3 Supervised Regression Method Bucket-Wise Details

The main data-requirements section reports the weighted supervised regression method comparison because it is easier to interpret in the main text. Tables B.7 and B.8 give the corresponding bucket-wise normalized residual standard deviation for the original bucket-wise comparison methods (XGBoost, HistGradientBoosting, Bagged trees, Random forest, Wide NN, Ridge). The later MLP sensitivity study is summarized separately in the main text because its bucket-specific and all-bucket training scopes were evaluated across the ten primary buckets. These tables are used to check whether the selected method is supported by consistent bucket-level behavior rather than only by the weighted aggregate.

Table B.7: $M1$ bucket-wise 10 s-window normalized residual standard deviation by model family.

Slope (%)	Temperature (°C)	XGBoost	Hist. GB	Bagged trees	Random forest	Wide NN	Ridge
-5 to -1	-10 to 0	3.28	3.28	3.65	3.22	3.34	3.87
1 to 5	-10 to 0	2.79	3.13	2.86	3.52	2.65	4.34
1 to 5	0 to 10	2.53	2.99	2.60	3.64	2.18	5.43

Table B.8: $M2$ bucket-wise 10 s-window normalized residual standard deviation by model family.

Slope (%)	Temperature (°C)	XGBoost	Hist. GB	Bagged trees	Random forest	Wide NN	Ridge
-5 to -1	-10 to 0	2.86	2.90	2.99	2.88	2.95	3.95
1 to 5	-10 to 0	1.67	1.70	1.70	1.71	2.01	3.13
1 to 5	0 to 10	1.62	1.61	1.64	1.73	1.77	4.24

Appendix C

Driving-Action Supporting Results

C.1 Detailed Driving-Action Diagnostics

Figure C.1 and Table C.1 show how $M3$ prediction error changes with target horizon. These details support the interpretation that the 1 s target is the most relevant immediate-action target, while longer point-ahead targets are mainly diagnostic because they assume recorded future context. All learned-model results in this section use the final XGBoost-based $M3$ model; persistence is included only as a baseline comparator.

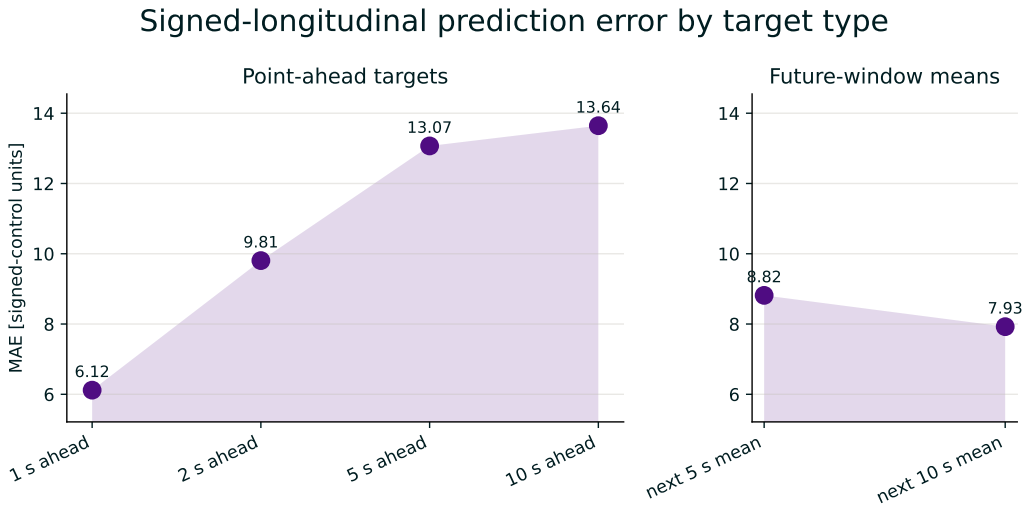


Figure C.1: Signed-longitudinal MAE by prediction horizon.

Table C.1: Direct $M3$ prediction metrics for the main primary buckets.

Prediction target	Buckets	Test rows	MAE	RMSE	R^2	Corr.
1 s ahead	4	373189	6.12	9.80	0.83	0.91
10 s ahead	4	373189	13.64	18.39	0.61	0.78
next 10 s mean	4	373189	7.93	10.66	0.74	0.86

Table C.2 compares $M3$ with the persistence baseline across the same target horizons. It shows

why persistence is a strong immediate-action baseline and why the $M3$ advantage is clearer at longer horizons.

Table C.2: $M3$ prediction compared with a persistence baseline.

Prediction target	Buckets	Rows	$M3$ RMSE	Persistence RMSE	$M3$ MAE	Persistence MAE	$M3$ R^2	Persistence R^2	Δ RMSE	Δ MAE	Δ R^2
1 s ahead	4	373189	9.80	11.62	6.12	6.36	0.83	0.76	-1.82	-0.24	0.07
10 s ahead	4	373189	18.39	36.13	13.64	25.89	0.61	-0.50	-17.74	-12.25	1.11
next 10 s mean	4	373189	10.66	24.13	7.93	17.06	0.74	-0.31	-13.47	-9.14	1.05

Figure C.2 and Table C.3 give the horizon-level feature-contribution details for $M3$. The main text reports the headline result; these diagnostics show how the preview and context contribution grows at longer prediction horizons.

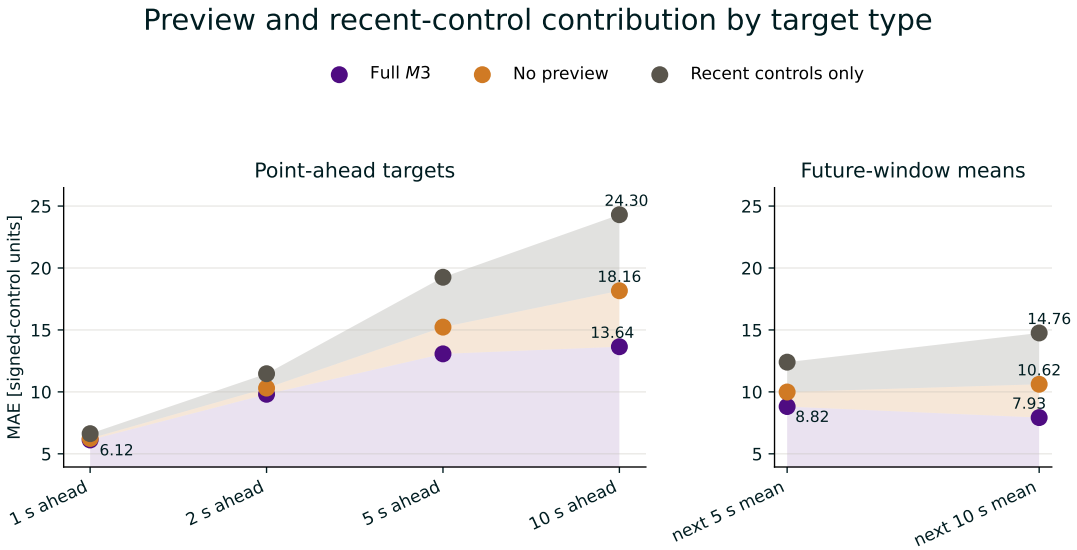


Figure C.2: Signed-longitudinal MAE by prediction horizon for the full $M3$ model, the no-preview variant, and the recent-controls-only variant.

C.2 $M3$-via-$M2$ Counterfactual Diagnostic

The $M3$-via-$M2$ counterfactual is retained as an earlier supporting diagnostic, not as primary RQ3 evidence. It replaces the realized driver-control inputs with $M3$-predicted inputs in the $M2$ feature representation while keeping speed, position, and road context fixed. It therefore measures the change predicted by $M2$ under altered controls, not the closed-loop energy effect of following $M3$ in a simulator or vehicle. Both models use XGBoost, and the analysis considers the immediate 1 s action only.

Define ΔEC as $M2$ with $M3$-implied controls minus $M2$ with realized controls. Under the thesis sign convention, positive ΔEC is favorable because it represents less battery drain or more charging. Figure C.3 and the accompanying Table C.4 report the mean ΔEC and the fraction of positive rows for each $M3$ bucket. The result motivated the subsequent shift toward direct prediction and simulator-based evaluation.

Table C.3: Preview and recent-control ablation for signed-longitudinal action prediction. All variants are evaluated over the four primary $M3$ buckets.

Variant	Prediction target	MAE	Δ MAE from full	RMSE	Δ RMSE from full
Full $M3$	1 s ahead	6.12	0.00	9.80	0.00
No preview	1 s ahead	6.24	0.12	10.06	0.25
Recent controls only	1 s ahead	6.63	0.51	10.97	1.17
Full $M3$	10 s ahead	13.64	0.00	18.39	0.00
No preview	10 s ahead	18.16	4.52	23.49	5.10
Recent controls only	10 s ahead	24.30	10.66	29.09	10.70
Full $M3$	next 10 s mean	7.93	0.00	10.66	0.00
No preview	next 10 s mean	10.62	2.69	14.20	3.53
Recent controls only	next 10 s mean	14.76	6.83	18.73	8.07

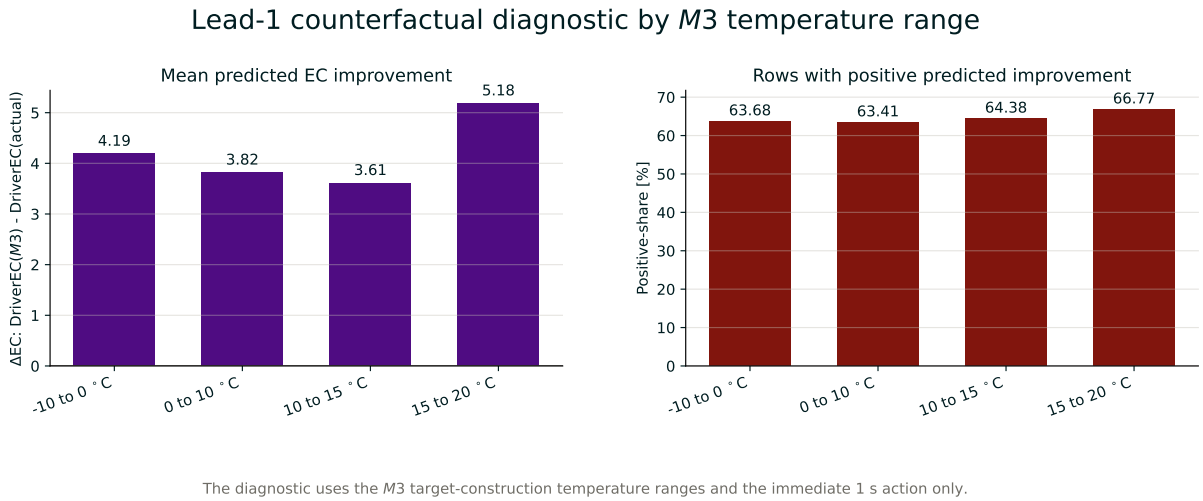


Figure C.3: Lead-1 $M3$-via-$M2$ counterfactual diagnostic by $M3$ bucket.

Table C.4: Lead-1 $M3$ temperature-range counterfactual diagnostic, sorted by mean predicted EC improvement. Delta EC is DriverEC with $M3$-implied controls minus DriverEC with realized controls.

Temperature (°C)	Mass [t]	Rows [M]	Mean ΔEC	P10 ΔEC	Median ΔEC	P90 ΔEC	Positive share [%]
15 to 20	17-21	0.27	5.18	-4.65	2.13	18.46	66.77
-10 to 0	17-21	1.72	4.19	-5.36	1.80	16.52	63.68
0 to 10	17-21	2.24	3.82	-5.39	1.66	15.41	63.41
10 to 15	17-21	1.01	3.61	-5.26	1.54	14.92	64.38

Figure C.3 and Table C.4 show a positive lead-1 diagnostic result in every $M3$ bucket. The $M3$ grouping is used because the purpose is to assess the driving-action model; $M2$ serves only as the fixed energy scorer.

The subset diagnostic (Figure C.4 and Table C.5) compares rows included in the $M3$ target-data population with the remaining processed rows. The model-based improvement is larger for the remaining rows and close to zero for the target-data subset, which is consistent with the selected subset already representing comparatively efficient controls. Comparisons against measured EC are retained only for calibration because the two subsets have different relationships between $M2$ predictions and measured energy consumption.

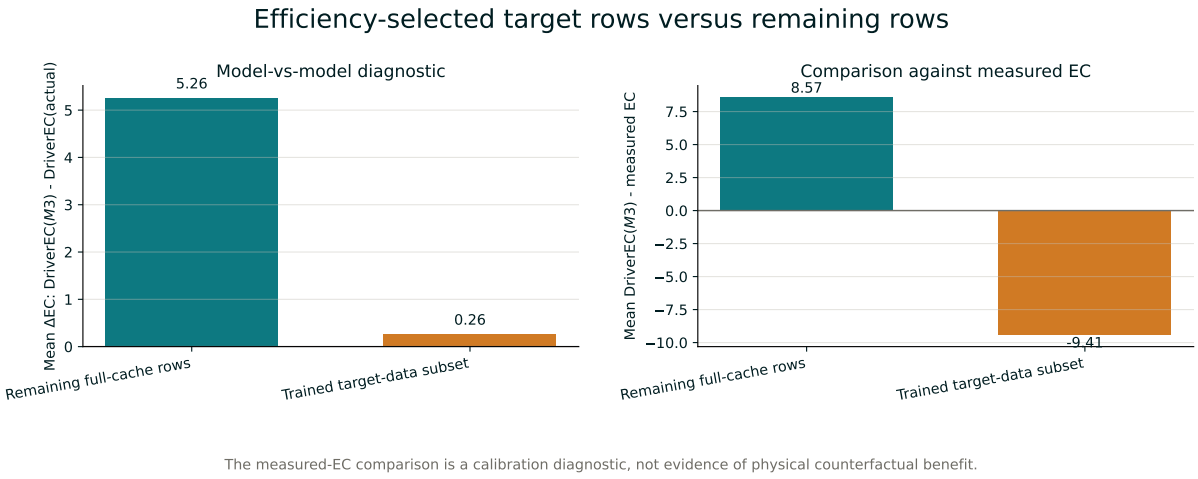


Figure C.4: Lead-1 $M3$-via-$M2$ counterfactual subset diagnostic.

Table C.5: Subset diagnostic for the $M3$-via-$M2$ counterfactual analysis. The target-data subset contains rows used to construct the $M3$ target data; the remaining rows are the rest of the full-cache lead-1 counterfactual rows.

Metric	Remaining full-cache rows	Trained target-data subset
Rows [M]	3.89	1.35
Products	31	30
Chunks	1185	1170
Mean $M2(M3)$ - $M2$(actual)	5.26	0.26
Median $M2(M3)$ - $M2$(actual)	2.98	-0.01
Mean $M2(M3)$ - measured EC	8.57	-9.41
Positive vs $M2$(actual) [%]	69.07	48.89
Positive vs measured EC [%]	71.86	20.33

C.3 Additional Recorded-Route Replay Evidence

This section adds a second recorded-route replay case as supporting evidence for the $M3$ control pattern described in Section 4.3.3.1. The full-range view and two zoomed segments show that $M3$ generally applies smoother accelerator variation and avoids the braking event seen in replay over the compared segments.

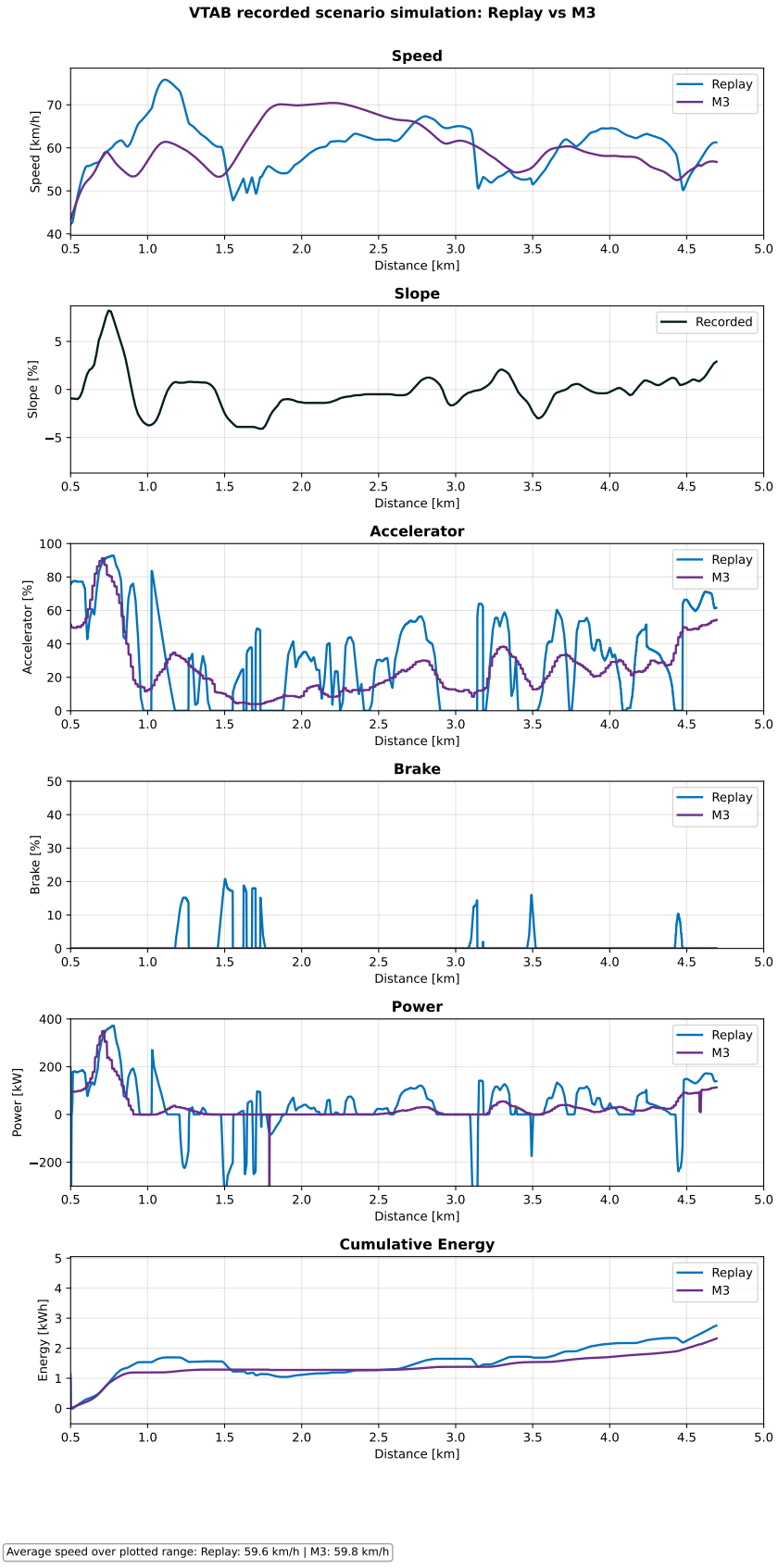


Figure C.5: Second recorded-route replay case: speed, slope, accelerator, brake, power, and cumulative energy for replay and $M3$.

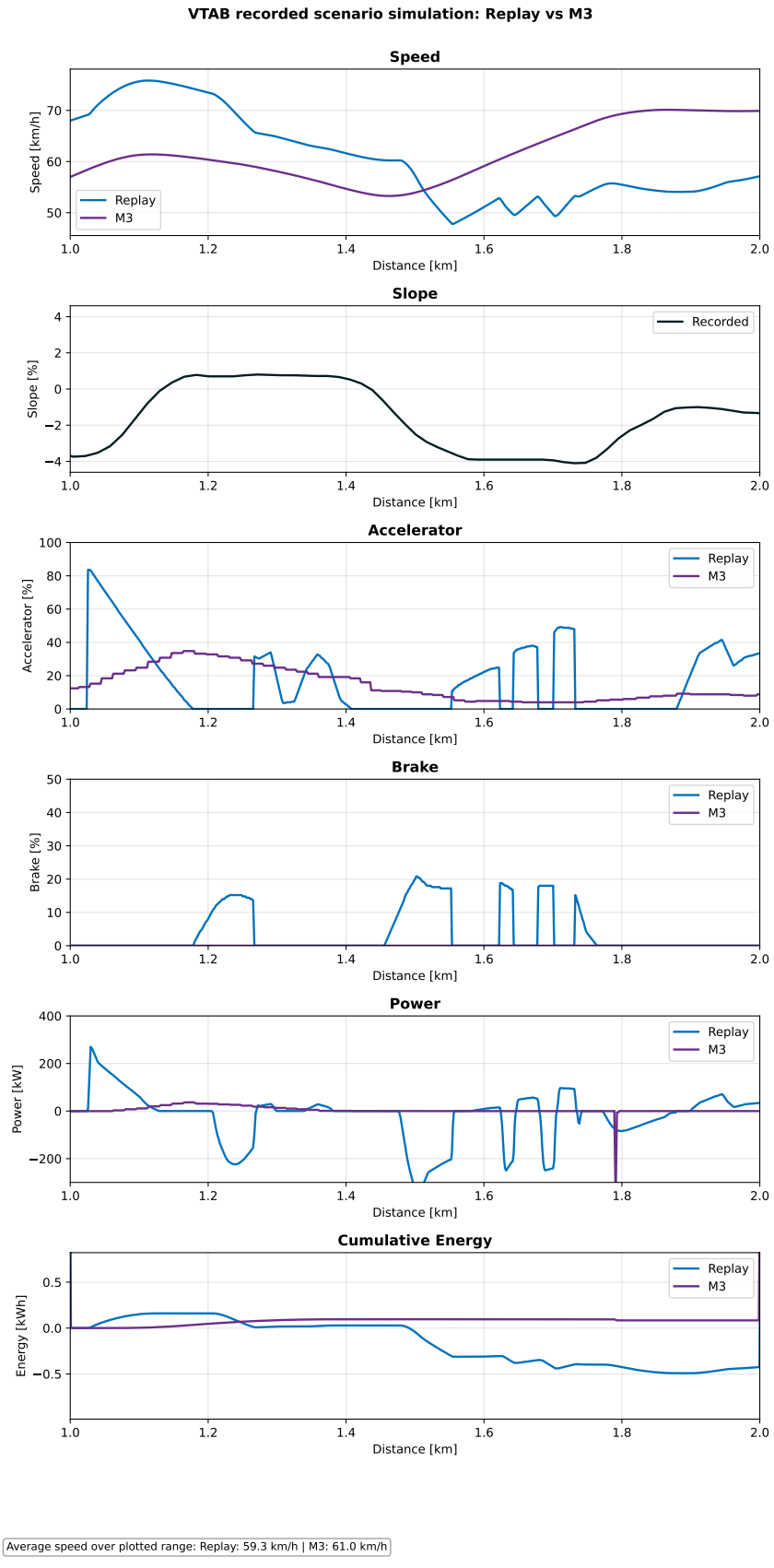


Figure C.6: Second recorded-route replay case, zoomed segment 1 (1.0–2.0 km).

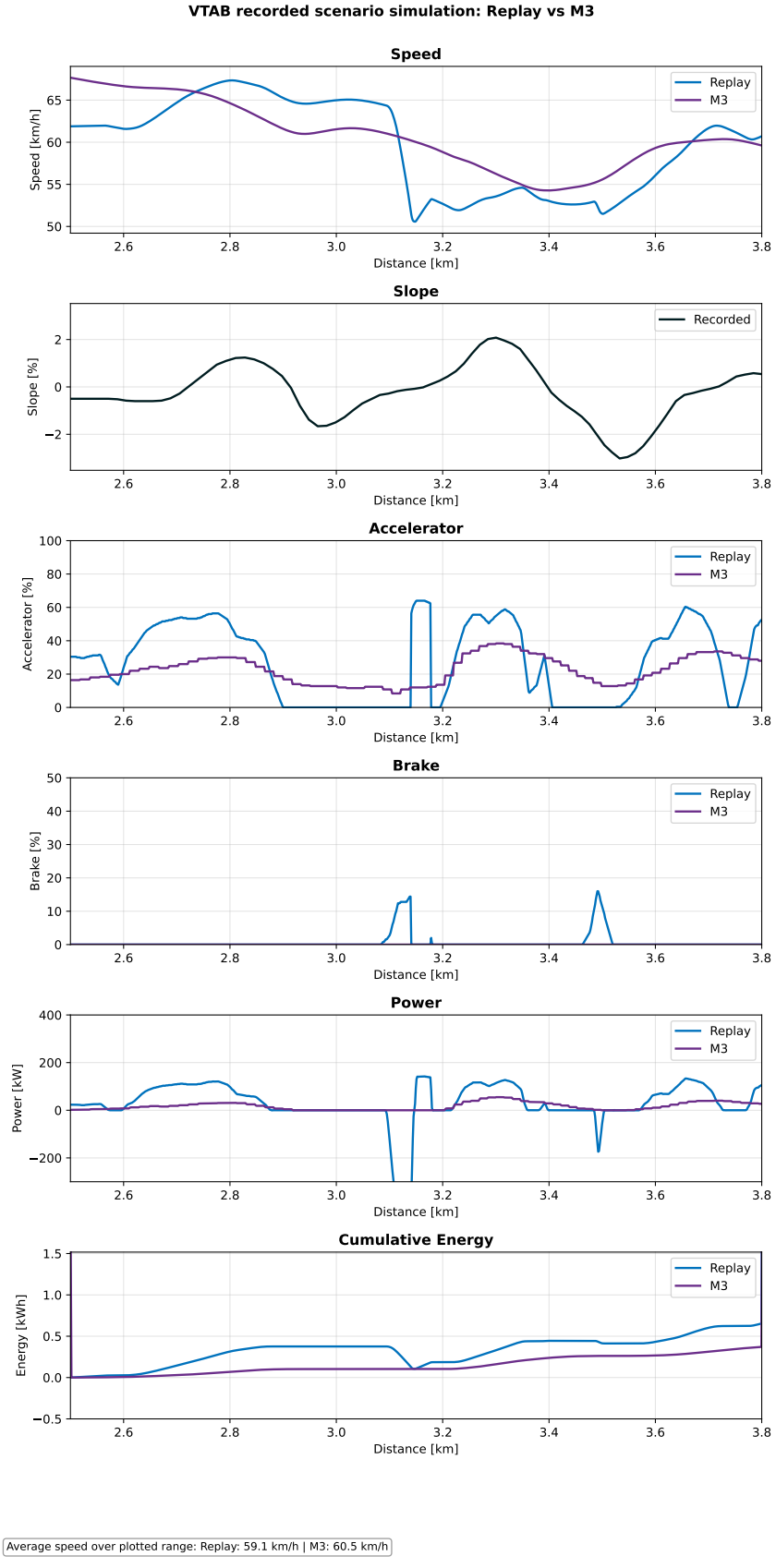


Figure C.7: Second recorded-route replay case, zoomed segment 2 (2.5–3.8 km).

Appendix D

Use of Generative AI

This appendix reports how generative AI (GenAI) tools were used in this thesis project for both technical work and report writing.

OpenAI Codex, including Codex CLI and GPT-family assistant models, was used for programming support and command construction. Typical use covered local code or log snippets, workflow questions, and summarising the results for inclusion in the thesis report. All model outputs and results used in the thesis were produced by the implemented and verified experiment pipeline, not by GenAI tools. OpenAI ChatGPT (GPT-family models) was used for explorative literature study, alternative wording proposals, and grammar check in the report.

In all cases, generated output was treated as draft support material and was manually revised by the author before integration. Technical claims were verified, and code and analysis behavior were validated before use in the thesis. Generative AI was not used to produce complete report sections without substantive author rewriting. Responsibility for scientific reasoning, implementation decisions, experiment validation, and final conclusions remains with the author.